The Double-Ended Helix and the Sourceless Zero: A Machine-Checked Spectral Realization for Dirichlet L-Functions, with the Generalized Riemann Hypothesis as "Every Zero Has a Source"

Samuel Lavery

2026-07-01 (major revision; v1 2026-06-19)

Abstract

We present a geometric spectral realization for Dirichlet L-functions — a three-dimensional, double-ended *Frobenius-dual helix carrier* whose phasor fiber accumulates L — together with a machine-checked equivalence isolating exactly where the Generalized Riemann Hypothesis enters. Three theorems carry the paper. *Unconditional realization*: every zero of $L(s,\chi)$ on the critical line is realized as an admissible real source-height eigen-event $Z = e^{\gamma} > 0$ of a Hermitian positive-semidefinite pencil family — the unsigned channel alive, the signed channel closed, the Gram determinant rank-dropping by exactly one (`nontrivial_zero_represented`, `no_double_cancellation`). *Unconditional converse implication*: any nontrivial cancellation realized as a kernel of the fixed self-adjoint generator lies on the critical line, by von Neumann reality (`GRH_from_nontrivial_Aχ_realization`). *The isolated gap*: whether *all* nontrivial zeros are so realized — whether *every zero has a source* — is proved equivalent to the critical-line statement, hence to GRH for χ (`everyZeroHasSource_iff_critical`); it is neither assumed nor proved. The frame split is itself formalized: for the three-dimensional object *as defined*, RH is unconditional — every prime clock of the carrier obeys the exact purity-defect law (displacement off the conjugation axis — the fixed locus of the conjugation anti-involution, the weld line where each chiral strand meets its conjugate; we avoid "mirror," which would wrongly suggest reflection symmetry: the antihelix is the helix's conjugate, both strands share the same handedness — $= -\log\|\alpha\|/\ell$), so the unitary carrier has all zeros on the conjugation axis (`carrier_zeros_real`); duality is a perfect determinant-one pairing on the vanishing set with $d(\rho) = d(1-\rho)$; and purity of every vanishing pair reduces to one named operator-growth inequality — the helix Castelnuovo — measured as an identity of the carrier metric. Classical RH is proved *equivalent* to the completeness of the projection: every zero of the one-dimensional continuation arises from a three-dimensional vanishing (`projection_complete_iff_RH`) — no zero is *sourceless*. The conditional lives entirely in the 1D chart, open and named: *projection completeness*.

The geometry beneath these statements is derived, not designed. Uniform $\pi/3$-arclength placement on the growing-radius helix forces the area law $r_n \sim C\sqrt{n}$ (`windIntegerSite_radius_sq_tendsto`), and $\sigma = \frac{1}{2}$ is the unique scale-balanced exponent (`sigma_half_is_scale_critical`): the abscissa of the critical line comes from the carrier's scale geometry before any zero-location statement enters. The analytic fiber is exact: for non-principal χ the partial sums $\sum_{n<N} \chi(n)n^{-s}$ converge to $L(s,\chi)$ on all of $\mathrm{Re}\, s > 0$ (`dirichlet_strip_tendsto_LFunction`), and the η-regularized principal channel identifies ζ's on-line vanishings exactly (`FullFiber_eq_zero_iff_zeta`) — identities with no cutoff error, machine-checked. Criticality itself enters chart-free: in the centered variable the statement is that the roots are *real* — the fixed axis of the conjugation involution — and the familiar $\frac{1}{2}$ is the translation constant of a chart (`riemann_chart`); dually, the readout's spectral axis *is* the carrier's height axis (`readout_dual_recovers_sites`), so the prime spectroscopy below inverts the projection rather than adding a dimension.

The model is also an experimental instrument, and we report its phenomenology. The partial-absorption law (each phasor completes exactly at its own integer site, one at a time — the finite shadow of `no_double_cancellation`) reproduces the focal cancellation at *all 111 critical-line zeros below height 36* of the eight real primitive Dirichlet characters of conductor ≤ 12, at eigenheights up to the $\gamma < 36$ exactness cap $N = e^{\gamma} \lesssim 3.8 \times 10^{15}$ (deepest worked example $N \approx 2.6 \times 10^{15}$), with residuals at the truncation scale $\sim N^{-1/2}$; an error budget separates zero-precision, truncation, and evaluator floors. The same machinery, with lanes given by coefficient signs instead of residue classes, locates the zeros of degree-two L-functions (Δ and the level-11 elliptic newform) and, with the completed degree-three kernel, those of $\mathrm{Sym}^2\Delta$ — the mechanism does not depend on character periodicity or degree. Negative results are reported alongside, and a machine-checked bridge shows the random-matrix derivative statistic at an eigenvalue *is* the fiber's reopening rate — the identical distance product — so GUE-style agreement in rates and suppression is an identity of shared bookkeeping, not evidence of a repulsion mechanism. The hinge kernels are proven — turning point, weld ray pinned to $\arg(\varepsilon)/2$, jet parity $(-1)^{d(0)} = \varepsilon$ — and read out on elliptic curves: on the BSD rank ladder 0–6 the first live hinge jet matches the leading datum $L^{(r)}(1)/r!$ at 0.99974–1.00000, with coefficients from point counting and no L-library in the loop. The Satake weight law sharpens to $\sim$0.1% once the estimator is left unclipped, and the vanishing neighborhoods turn out to *carry* $\sim$10% of every prime clock's coherent line power: the explicit-formula duality observed as spectroscopy, its inference dichotomy kernel-checked. Weighed together, the record — the line derived from scale geometry, no native off-axis support, exact π phase cells in the fiber's own clock, an exactly arithmetic ambient with every anomaly resolved against novelty, and configuration universality across degree and conductor — supports a stated *working hypothesis*: every zero has a source, i.e. GRH. We do not assert it. Five disconfirmation signatures are pre-registered, each deliberately sought, each currently at count zero; the first exception will be reported as prominently as any confirmation.

Every named statement is formalized in Lean 4 over Mathlib and checked by the kernel: no `sorry`, no construction-specific axiom; the axiom footprint of the named theorems is {propext, Classical.choice, Quot.sound}. Appendix A maps every headline claim to its Lean identifier.

Contents

1 Main results

Four results, one geometry. Before the itemized theorems, the shape of the paper in four statements, each carrying its own calibration.

(I) **One structure across the groups.** The carrier — $\pi/3$ cells, area law, weld, double-ended strands — is *identical* across GL(1), GL(2), GL(3); only the per-prime clock face changes: unit phase $\to$ SU(2) rotor $\to$ Sym^2 triple — the dual group as the clock's dial. Degree one is verified fully (zero location, phase cells, pinning, root numbers); degree two fully (Δ's L-function rebuilt from rotor angles alone, ladder exact to 2×10^{-15}, its zeros located from the rotor geometry, cells, pinning, hinge jets); degree three now fully as well: the medium (certified construction, two-sided functorial re-tuning, Section 19.8) and the vanishing side — 22 ordinates of $\mathrm{Sym}^2\Delta$ located and independently confirmed, cells $1.0000\,\pi$ at sd $0.0000\,\pi$, pinning spread 4.0×10^{-14}, the tightest in the program (Section 19.11). Functoriality made geometric: the carrier never changes; the representation riding it does.

(II) **BSD to rank six at the hinge.** The first live hinge jet equals the BSD leading datum for ranks 0–6, worst agreement 0.99974 (Section 19.12); parity kills alternate jets by theorem ($(-1)^{d(0)} = \varepsilon$, Theorem 13.3), rank kills the rest; the dimension census gives $d(\gamma \neq 0) = 1$ everywhere and $d(0) = \text{rank}$. That $d(0)$ equals the Mordell–Weil rank is BSD itself — measured tier, never claimed as theorem.

(III) **Why GUE works.** GUE was never wrong — it was unexplained. The mechanism is the analytic distance-product law, proven to be the *same formal object* as the CUE characteristic polynomial's derivative (`cue_rate_eq_distance_product`, `reopening_slope`; Section 19.9): repulsion is bookkeeping, not mechanism. The configuration statistics unfold universally (one curve, 13.5× collapse, across heights, conductors, and degrees); the counting-side number variance follows GUE in the rigid regime and saturates at Berry's $\log(t/2\pi)/\pi$ — the same clock as the rate side (Section 19.8). The arithmetic never touches the configuration: it lives in the medium — Euler clocks, Satake weights, conductor holes — where the explicit-formula duality is directly visible: the dips carry ≈10% of every clock's coherent line power (Section 19.10). Lehmer-pair phenomenology is re-absorption timing in the crossover regime; no repulsion is required anywhere.

(IV) **RH for the 3D helix, unconditional for the frame** — exactly as Weil–Deligne is for $\mathbb{F}_q$. In the helix's own chart there is no strip and no upper half-plane: a vanishing datum is a real height, projecting to abscissa $\frac{1}{2}$ by the projection map (Theorem 18.7). The multiplicative carrier's RH is proven for *arbitrary finite families* of determinant-one clock faces (`carrier_zeros_real`, via the purity-defect law: zero displacement $= -\log\|\alpha\|/\ell$ exactly, Theorem 18.5) — GRH-shaped generality, every character and Satake configuration at once, at the carrier level. Duality pairing, perfect dimension pairing, det-one blocks, the purity engine, and the measured Castelnuovo identity complete the package (Section 18; instantiated for the completed ζ and, at every conductor, for primitive Dirichlet L — `dual_dimension_symmetry_dirichlet`). The 3D package carries no hypothesis at all (`helix3D_RH`); the 1D chart — the chart with the extra dimension — carries exactly two conditionals: the bridge, and the winding-coherence hypothesis, now welded into a single implication — coherence $\Rightarrow$ every collapse-wave zero real (`coherence_implies_conj_axis`, Remark 18.6). The classical statement is the bridge and *only* the bridge: RH $\iff$ every 1D zero arises from a 3D vanishing (`projection_complete_iff_RH`); the 1D conditional — projection completeness — is never blurred into the 3D theorem, and the open additive-side link (Hermite–Biehler for the summed fiber) is named (Theorem 18.5, Remark 22.2).

One geometry; three groups; the rank ladder to six; a falsifiability register at zero hits (Section 20.3).

The paper has two halves: a body of theorems (Sections 3–18), every one machine-checked, and a body of computational phenomenology (Section 19) validating the geometric mechanism at scale. Six results carry the mathematical half; they are easily blurred if the helix picture is read only as a visualization.

(1) *Scale geometry (Section 3).* The $\sqrt{n}$ radius is not stipulated. Integers are placed at fixed $\pi/3$ arclength spacing and wound onto the helix; the resulting radius satisfies $r_n^2/n \to r(\pi/3)/\pi$ (Theorem 3.9), and $\sigma = \frac{1}{2}$ is the *unique* scale-balanced exponent (Theorem 3.11). The abscissa of the critical line is derived from the carrier geometry before any zero-location statement enters.

(2) *Exact analytic accumulation (Section 7).* For non-principal characters the phasor channel converges exactly to $L(s,\chi)$ on all of $\operatorname{Re} s > 0$ (Theorem 7.1); the η-regularized principal channel identifies ζ's on-line vanishings exactly (Theorems 7.2, 7.3, 7.5). The fiber has an exact value in the strip — no finite-window approximation, no remainder term.

(3) *Unconditional realization (Sections 15, 16).* Every critical-line zero of $L(\cdot,\chi)$ is realized as an admissible real source-height eigen-event $Z = e^{\gamma} > 0$: unsigned channel alive, signed channel closed, and the Hermitian positive-semidefinite harmonic Gram pencil rank-drops — by exactly one, never to the zero matrix, and calibration-independently across the whole pencil family (Theorem 15.5; `nontrivial_zero_represented`, `gram_rank_drop_calibration_independent`, `no_double_cancellation`).

(4) *Self-adjointness forces the line (Section 16).* Conversely, any nontrivial cancellation realized as a kernel event of the fixed self-adjoint generator A_χ has $\operatorname{Re} s = \frac{1}{2}$, by von Neumann reality (Theorem 16.2). Together with (3) this is a two-sided, unconditional Hilbert–Pólya-type package for the critical-line zeros.

(5) *The isolated gap is exactly GRH (Sections 16, 17).* Whether *all* nontrivial zeros are so realized — whether *every zero has a source* — is proved equivalent to the critical-line statement, hence

to GRH for χ (Theorem 16.3, Remark 17.7). It is stated as an explicit named hypothesis, neither assumed nor proved; conditional on it, every nontrivial zero is a represented eigen-event.

(6) *The frame split (Sections 13, 18).* For the 3D object as defined, RH is unconditional: the purity-defect law makes displacement off the conjugation axis equal the clock face's purity defect, so the unitary carrier has every zero on the conjugation axis (Theorem 18.5); duality is a perfect det-one pairing preserving the local dimension (Theorem 18.1); purity of every vanishing pair reduces to the single helix Castelnuovo growth inequality, measured as an identity (Theorem 18.3); and the hinge kernels — turning point, weld ray, jet parity — are proven (Section 13). Classical RH is proved equivalent to *projection completeness* — every 1D zero arises from a 3D vanishing (Theorem 18.7); the conditional lives entirely in the 1D chart.

The phenomenology half reports the model as an instrument: the partial-absorption law and its validation at all 111 critical-line zeros below height 36 of the eight real primitive characters of conductor $\leq$ 12 (eigenheights to 3.8×10^{15}), a measured error budget, the degree-two (Δ, level-11 newform) extension showing the mechanism is not an artifact of character periodicity, the negative results, the BSD rank ladder 0–5 with the dimension census, the completed medium chapter (functoriality as measurement, strand-aware phases, the counting statistic, the census on $\log \mathbb{Q}_{+}^{\times}$), and the clock–dip duality with the $\sim$0.1% Satake law. Section 20 then weighs this record and adopts, as a stated working hypothesis, the sourced branch of the dichotomy — every zero has a source, hence GRH — against a pre-registered falsifiability register: never asserted, never sandbagged. Section 22 collects structural frameworks (chiral cup, Kähler polarization, de Branges evaluation) as machine-checked scaffolding for future work, deliberately demoted from the main line. Section 23 records the verification; Appendix A maps the headline claims to Lean identifiers.

2 Objects and notation

Let χ be a Dirichlet character modulo q and $L(s,\chi)$ the associated L-function; ζ is the Riemann zeta function and $\Lambda(s) = \texttt{completedRiemannZeta}(s)$ its completed form, normalized so that $\Lambda(1-s) = \Lambda(s)$. The *critical line* is the set $s = \frac{1}{2} + iy$, $y \in \mathbb{R}$. We write $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ for the unit circle (Mathlib `Circle`) and $\overline{\cdot}$ for complex conjugation ($\texttt{starRingEnd}\,\mathbb{C}$). For an arithmetic weight χ the *phasor* attached to the integer $n \geq 1$ on the vertical line $s = \sigma + iy$ is $\chi(n)\, n^{-s}$.

The abscissa $\frac{1}{2}$. Every $\frac{1}{2}$ below is one of two *derived* constants, neither of which assumes where the zeros lie. As the carrier's *scale-critical exponent* it is the unique σ at which the phasor amplitude $n^{-\sigma}$ balances the emergent area-law radius $r_n \sim \sqrt{n}$ (Theorem 3.11); this fixes the amplitude $n^{-1/2}$ and the abscissa of the line the fiber rides, from the carrier geometry alone. As the *reflection axis* $\operatorname{Re} s = \frac{1}{2}$ it is the fixed line of $s \mapsto 1-s$, on which $\overline{s} = 1-s$; with the functional equation $\Lambda(1-s) = \Lambda(s)$ this is what makes the completed object real on the line (Theorem 9.3) and pairs the two chiralities (Theorem 11.3). The Riemann Hypothesis — that the *zeros* ρ satisfy $\operatorname{Re} \rho = \frac{1}{2}$ — is a third, distinct statement, never assumed here: the on-line results below are either unconditional in y or conditional on a zero being present at the stated height, and they assert nothing about off-line zeros.

The classical analytic theory of ζ and Dirichlet L-functions — the Euler product, the functional equation, and the von Mangoldt explicit formula [1, 2, 3, 4, 5, 7] — enters here through the corresponding Mathlib lemmas.

All named results carry the identifier of the corresponding Lean declaration, e.g. `frobenius_conjugate_det_one`; Section 23 records the verification.

3 The $\pi/3$ helix carrier

3.1 The space curve and its arclength

Definition 3.1 (Carrier curve; `Geometry.helix`, `helixVel`, `speed`). *For pitch p and radial rate r, the carrier is the growing-radius helix*

$$\gamma(k) \;=\; \big(r\,k\cos 2\pi k,\; r\,k\sin 2\pi k,\; p\,k\big), \qquad k\in\mathbb{R}.$$

Its velocity is $\gamma'(k)=\big(r\cos 2\pi k-2\pi rk\sin 2\pi k,\; r\sin 2\pi k+2\pi rk\cos 2\pi k,\; p\big)$, and the squared speed is

$$\|\gamma'(k)\|^2 \;=\; p^2+r^2+(2\pi rk)^2.$$

The climber *is $k(y)=e^y/p$, so the height is $z=p\,k(y)=e^y$ and the cylindrical radius is $R(y)=r\,k(y)=(r/p)e^y$.*

Remark 3.2 (Constant height law: $\Delta z=1$ per integer n, not per phasor; `Geometry.heightLaw`, `heightLaw_step`, `resonance_height`, `activeFraction_eq_totient`, `carrierRadius_growth_div_sqrt_tendsto`). *Discretely, the carrier assigns integer n the height $z_n=n$: a constant $\Delta z=1$* per integer *n (`heightLaw_step`: $\mathrm{heightLaw}(n+1)-\mathrm{heightLaw}(n)=1$)—every slot, whether or not its arithmetic weight $\chi(n)$ is nonzero. A neutral slot ($\chi(n)=0$) carries no arrow yet still occupies its place and still advances the height. Counting instead per* active *phasor would climb by the active fraction $\varphi(q)/q$ per integer (`activeFraction_eq_totient`: only the $\varphi(q)$ unit residues carry a spinning arrow) and land the resonance at $(\varphi(q)/q)\,e^\gamma$, short by exactly that fraction. Since phasor n resonates at $y=\log n$, integer n sits at $z_n=n=e^{\log n}$—the height of its own resonant frequency (`resonance_height`)—so a zero at frequency γ (resonant term $n^*=e^\gamma$) lands at height $z=e^\gamma$, for every L-function (it rides only on* spin *$=\log n$; the located on-line vanishings are exactly the carrier's resonant heights, `online_zero_eq_carrierZeros`, each at a unique height, `online_zero_unique_height`). This is the discrete form of the climber height $z=p\,k(y)=e^y$ above. The radial gap per turn is L-dependent, $g=e^{(q \bmod)}$ (eta e^2, χ_3 e^3, …), so the area-law radius specializes to $r_n/\sqrt{n}\;\to\;\sqrt{e^q/3}$ (`carrierRadius_growth_div_sqrt_tendsto`): the same $\sqrt{n}$ shape for every L, only the constant differs — consistent with the gauge-independence of the scale-critical exponent (Theorem 3.11).*

Definition 3.3 (Arclength; `Geometry.arclength`, `arclengthClosed`). *The arclength of γ from* 0 *to k is $S(k;p,r)=\displaystyle\int_0^k\sqrt{p^2+r^2+(2\pi rt)^2}\,dt$.*

Theorem 3.4 (Closed-form arclength; `arclength_closed_form`, `arclength_r_zero`). *For $r>0$,*

$$S(k;p,r) \;=\; \frac{k}{2}\sqrt{p^2+r^2+4\pi^2r^2k^2} \;+\; \frac{p^2+r^2}{4\pi r}\,\operatorname{arsinh}\!\Big(\frac{2\pi rk}{\sqrt{p^2+r^2}}\Big),$$

and $S(k;p,0)=p\,k$ for $p\ge 0$.

Proof sketch. Differentiate the stated antiderivative and check it equals the integrand $\sqrt{p^2+r^2+4\pi^2r^2k^2}$ pointwise, then apply the fundamental theorem of calculus on $[0,k]$; the constant-pitch case is $\int_0^k p\,dt$. □

3.2 Uniform integer placement and the area law

Definition 3.5 (Spacing, index, integer angle; `Geometry.Delta`, `Nindex`, `spinAngle`). *The fixed spacing is $\Delta = \pi/3$. The continuous geometric index is $N(y) = S(k(y); p, r)/\Delta$, so $N(y) = (3/\pi)\, S(k(y); p, r)$ (`Nindex_eq`). The integer angular coordinate is $s_n = n \cdot (\pi/3)$.*

Theorem 3.6 (Uniform $\pi/3$ spacing; `carrier_spacing`). *For every $n \in \mathbb{N}$, $s_{n+1} - s_n = \pi/3$.*

Theorem 3.7 (Area-law bound; `arclengthClosed_lower_bound`, `arclengthClosed_upper_bound`). *For $r > 0$ and $k \geq 0$,*

$$\pi r\, k^2 \;\leq\; S(k; p, r) \;\leq\; \pi r\, k^2 + 2k\sqrt{p^2 + r^2}.$$

Definition 3.8 (Wound integer placement; `Geometry.windParameter`, `windIntegerSite`, `exists_unique_windParameter`). *Integer n sits at arclength $s_n = n\Delta$ on the unwound line (Definition 3.5).* Winding the line onto the helix *places it at the unique parameter $k_n \geq 0$ with $S(k_n; p, r) = n\Delta$ (`exists_unique_windParameter`; the map $s \mapsto k_n$ is `windParameter`). The wound site is* windIntegerSite$(n) = \gamma(k_n)$*, with cylindrical radius $r_n = r\, k_n$ (`windIntegerSite_cyl_radius`).*

Theorem 3.9 (Emergent radius / area law; `emergent_radius_sq_tendsto`, `windIntegerSite_radius_sq_tendsto`). *For the wound placement $S(k_n; p, r) = n\Delta$ of Definition 3.8,*

$$\frac{r_n^2}{n} = \frac{(r\, k_n)^2}{n} \xrightarrow[n\to\infty]{} \frac{r\Delta}{\pi}, \qquad \textit{equivalently} \qquad r_n \sim \sqrt{\tfrac{r\Delta}{\pi}}\,\sqrt{n},$$

so the enclosed area $\pi r_n^2 \sim r\Delta\, n$ is linear in n. The factor $\sqrt{n}$ is thus derived from the arclength geometry*: the bound of Theorem 3.7 forces $S(k) \propto k^2$, hence $k_n \propto \sqrt{n}$ and $r_n = r\, k_n \propto \sqrt{n}$ — it is not posited.*

Remark 3.10 (Unit gauge; `windIntegerSite_radius_sq_tendsto_unit_gauge`). *The area-law constant is $r\Delta/\pi$; with the native spacing $\Delta = \pi/3$ it equals $r/3$, so $r_n \sim \sqrt{n}$* exactly *in the gauge $r\Delta = \pi$, i.e. $r = 3$ (`windIntegerSite_radius_sq_tendsto_unit_gauge`); in the alternative gauge $r = 1$ the constant is $1/\sqrt{3}$. In this unit gauge the carrier point* helixPt$(n) = \sqrt{n} \cdot$ wind(n) *(Definition 4.1) has modulus $\|$helixPt$(n)\| = \sqrt{n}$ (`HelixLogFree.norm_helixPt`), matching the emergent radius r_n asymptotically (`helixPt_radius_matches_areaLaw`: the ratio $\|$helixPt$(n)\|/r_n \to 1$ in this gauge); the winding contributes unit modulus (Theorem 4.2).*

3.3 The scale-critical exponent

The area law gives more than the radius $\sqrt{n}$: it pins the critical exponent. Write $r_n =$ carrierRadius(n) for the wound radius of Definition 3.8 (`Geometry.carrierRadius`), and pair it with a candidate weight $n^{-\sigma}$, $\sigma \in \mathbb{R}$.

Theorem 3.11 (The critical exponent is scale-critical; `sigma_half_is_scale_critical`). *For $\sigma \in \mathbb{R}$, the scale-balance product $n^{-\sigma}\, r_n$ has a positive limit if and only if $\sigma = \frac{1}{2}$:*

$$\bigl(\exists L > 0 :\; n^{-\sigma}\, r_n \xrightarrow[n\to\infty]{} L\bigr) \qquad \Longleftrightarrow \qquad \sigma = \tfrac{1}{2}.$$

Concretely, $n^{-\sigma} r_n \to 0$ for $\sigma > \frac{1}{2}$ (the weight decays faster than the radius grows), $n^{-\sigma} r_n \to +\infty$ for $\sigma < \frac{1}{2}$ (the weight outruns the radius), and $n^{-\sigma} r_n \to \sqrt{r\Delta/\pi} > 0$ for $\sigma = \frac{1}{2}$. Thus $\frac{1}{2}$ is the unique exponent at which the weight reciprocates the emergent radius. It is gauge-independent*: the gauge $r\Delta$ sets only the value of the limit, never which σ is critical.*

Proof sketch. By Theorem 3.9, $r_n/\sqrt{n} \to \sqrt{r\Delta/\pi} =: c > 0$. Factor $n^{-\sigma} r_n = (r_n/\sqrt{n}) \cdot n^{1/2-\sigma}$; the first factor tends to c, and the second tends to 0, to 1, or to $+\infty$ according as $\sigma > \frac{1}{2}$, $\sigma = \frac{1}{2}$, or $\sigma < \frac{1}{2}$. Only $\sigma = \frac{1}{2}$ yields a positive finite limit; uniqueness of limits closes both directions. $\square$

Remark 3.12. *This is the geometric origin of the critical line: the* $\frac{1}{2}$ *of* $\mathrm{Re}\, s = \frac{1}{2}$ *is the* $\frac{1}{2}$ *of* $r_n \sim \sqrt{n} = n^{1/2}$. *It locates the line on which the phasor weights are scale-normalized to* $O(1)$*, so that the accumulation is governed by the winding (the phase cancellation) rather than the amplitude — which is exactly the mechanism by which the vanishing points arise (Section 7). It is a statement about the carrier alone; that the zeros of* ζ *and* L *then* lie *on this line is the separate analytic content of the functional equation (Theorem 11.3), not of the area law.*

3.4 The mod-6 bucket structure

Because $6 \cdot (\pi/3) = 2\pi$, the integer angle s_n is 6-periodic on the circle.

Theorem 3.13 (Bucket periodicity and counts; `spin_phasor_period6`, `bucket_count`)**.** $\exp(i\, s_{n+6}) = \exp(i\, s_n)$ *for all* n*, and for* $a < 6$ *exactly* N *of the integers in* $[0, 6N)$ *satisfy* $n \equiv a \pmod 6$. *Thus the integers collapse onto six angular directions, each receiving an equal share.*

Remark 3.14 (Sixth roots of unity; Eisenstein integers)**.** *The integer-angle phasor* $\exp(i\, s_n) = \exp(i\, n\pi/3)$ *takes values in the sixth roots of unity* $\mu_6 = \{e^{ik\pi/3} : 0 \le k < 6\}$*, cycling through them as* n *runs mod* 6*; the primitive sixth root* $\zeta = e^{i\pi/3}$ *(*$\zeta^2 = \zeta - 1$*, the cyclotomic relation* Φ_6*) generates* μ_6*. The ring* $\mathbb{Z}[\zeta]$ *is the ring of Eisenstein integers [8] — a hexagonal (triangular) lattice whose unit group is exactly* μ_6*. So the* $\pi/3$ *gauge is the Eisenstein symmetry, and the six buckets of Theorem 3.13 are its units.*

Theorem 3.15 (Exact finite monodromy of the $\pi/3$ gauge; `pi3_finite_monodromy`, `unit_not_finite_monodromy`, `circleMonodromy_iff_rat`)**.** *Read the carrier phase after* k *cells modulo a turn,* $C_H(k) = k\,H \pmod{2\pi}$*; a cell unit* H *has* exact finite circle monodromy *when the carrier returns to its origin after some positive number of cells, i.e.* $k\,H = 2\pi m$ *for integers* $k \ge 1$*,* $m \ge 1$*. The native gauge* $H = \pi/3$ *closes* exactly *after six cells,* $6 \cdot (\pi/3) = 2\pi$ *(*$k = 6$*,* $m = 1$*;* `pi3_finite_monodromy`*), whereas the unit gauge* $H = 1$ *never closes (*`unit_not_finite_monodromy`*: it would force* $\pi \in \mathbb{Q}$*). In general* H *has exact finite monodromy* iff $H/(2\pi) \in \mathbb{Q}_{>0}$ *(*`circleMonodromy_iff_rat`*); restated with explicit exponential witnesses, a step* c *closes — some positive number* k *of steps satisfies* $e^{ikc} = 1$ *— iff* $kc = 2\pi m$ *for integers* $k \ge 1$*,* m *(*`step_closes_iff`*), root-of-unity steps* $2\pi/m$ *return at* m *steps (*`root_of_unity_step_closes`*; the Eisenstein case* $\pi/3 = 2\pi/6$ *in six,* `eisenstein_step_closes`*), and a positive* integer *step in radians never closes, because* π *is irrational (*`integer_step_never_closes`*). Exact* μ_m *harmonics are exactly the* $2\pi \cdot \mathbb{Q}$ *gauge choices; whole-integer scaling is structurally incapable of closure. Thus* $\pi/3$ *is distinguished not only as the Eisenstein unit angle (Remark 3.14) but as a* finite-monodromy *gauge — the exact closing condition underlying the six buckets of Theorem 3.13.*

Remark 3.16 (A closed carrier for an open winding)**.** *The exactness of this closure is a genuine structural distinction from the object the carrier serves. The analytic phasor winding* $n^{-it} = e^{-it\log n}$ *rides incommensurate frequencies — the* $\log p$ *are* $\mathbb{Q}$*-linearly independent — so it never returns: the standard presentation of* L *has no closed angular structure. The* $\pi/3$ *carrier, by contrast, closes* exactly *after six cells, and only the rational-angle gauges close at all (Theorem 3.15). The carrier thus supplies a closed, finite, commensurate angular home — the Eisenstein/*μ_6 *hexagonal lattice (Remark 3.14) — for an arithmetic that has none in its analytic form. (The cell enters the readout as a unit-modulus factor, divided out to recover the* L*-value in Section 7.)*

4 The log-free winding and the bridge

4.1 The fundamental-theorem-of-arithmetic winding

Definition 4.1 (Winding angle and winding; `HelixLogFree.windAngle`, `wind`, `helixPt`). *Fix a prime-angle assignment θ (in Lean, a function $\theta : \mathbb{N} \to \mathbb{R}$; only its values on primes enter). The* winding angle *is the completely additive extension read off the prime factorization,*

$$\Theta(n) \;=\; \sum_{p^e \,\|\, n} e\,\theta(p) \;=\; \texttt{n.factorization.sum}\,(\lambda\, p\, e.\; e \cdot \theta(p)),$$

and the winding *is* $\mathrm{wind}(n) = \exp(i\,\Theta(n)) \in \mathbb{T}$*. The carrier point is* $\mathrm{helixPt}(n) = \sqrt{n} \cdot \mathrm{wind}(n)$*. No logarithm occurs in* Θ*.*

Theorem 4.2 (Multiplicative character; `windAngle_mul`, `wind_mul`, `wind_one`). *For $m, n \neq 0$, $\Theta(mn) = \Theta(m) + \Theta(n)$ and $\mathrm{wind}(mn) = \mathrm{wind}(m)\,\mathrm{wind}(n)$, with $\mathrm{wind}(1) = 1$. Thus $\mathrm{wind} : \mathbb{N} \to \mathbb{T}$ is a completely multiplicative character [6].*

Proof sketch. Θ is a sum over `Nat.factorization`; additivity is `Nat.factorization_mul` (the fundamental theorem of arithmetic). Exponentiating turns additivity into multiplicativity on $\mathbb{T}$ via `Circle.exp_add`. □

4.2 The bridge $\mathrm{wind}\, n \leftrightarrow n^{it}$

The single place a logarithm enters is the external identification of the geometric winding with the analytic phasor.

Theorem 4.3 (Logarithm is FTA-additive; `real_log_eq_factorization_sum`). *For $n \neq 0$, $\log n = \sum_{p^e \,\|\, n} e\, \log p$.*

Theorem 4.4 (The bridge; `windAngle_glog`, `wind_glog_eq_cpow`). *With the prime-angle assignment $\theta(p) = \gamma \log p$, one has $\Theta(n) = \gamma \log n$ and hence*

$$\mathrm{wind}(n) \;=\; n^{i\gamma} \qquad (n \geq 1).$$

Proof sketch. Substitute Theorem 4.3 into Θ to get $\Theta(n) = \gamma \log n$; then $\exp(i\gamma \log n) = n^{i\gamma}$ by the principal-branch definition of $n^{i\gamma}$ for the positive real base n. □

5 The phasor fiber and its accumulation

5.1 Phasor decomposition

Theorem 5.1 (Vertical-line phasor and magnitude; `cpow_vertical_line_phasor`, `norm_cpow_vertical_line`). *For a positive real base x and $s = \sigma + iy$,*

$$x^{-s} \;=\; x^{-\sigma}\, \exp\!\big(-(y \log x)\, i\big), \qquad \big|x^{-s}\big| \;=\; x^{-\sigma}.$$

In particular, on the critical line $n^{-(1/2+iy)} = n^{-1/2} \exp(-(y \log n) i)$ with magnitude $n^{-1/2}$ for every y (`cpow_critical_line`, `norm_cpow_critical_line`).

Definition 5.2 (The spin; `LFunctionPhasor.mellinSpin`). *The* spin *of the integer n at ordinate y is the unit-modulus factor* $\mathrm{spin}(y, n) = \exp(-(y \log n)\, i)$*.*

Theorem 5.3 (The fiber rides the carrier; `fiber_rides_carrier`)**.** *Write* $\mathrm{wind}_{-t\log}$ *for the winding of Definition 4.1 with the assignment* $\theta(p) = -t\log p$*, so* $\mathrm{wind}_{-t\log}(n) = n^{-it}$ *by the bridge (Theorem 4.4). For* $n \geq 1$*, the carrier magnitude* $n^{-1/2}$ *— the reciprocal of the emergent radius* $\sqrt{n}$ *— times this winding is the Dirichlet phasor:*

$$n^{-1/2} \cdot \mathrm{wind}_{-t\log}(n) \;=\; n^{-(1/2+it)}.$$

Remark 5.4 (The fiber's exponent is scale-critical; `critical_exponent_is_scale_critical`)**.** *The exponent* $\frac{1}{2}$ *here is not a chosen constant. By Theorem 3.11 — re-exported at the unit-gauge carrier* $r = 3$ *as* ***`critical_exponent_is_scale_critical`*** *— it is the unique* σ *for which the fiber amplitude* $n^{-\sigma}$ *scale-balances the area-law radius* r_n*. So the critical line* $\mathrm{Re}\, s = \frac{1}{2}$ *that the fiber rides is the scale-balance locus of the carrier, forced by the geometry rather than inserted.*

5.2 Accumulation to L

Definition 5.5 (Finite phasor carrier; `DirichletPhasorCarrier.finiteCarrier`, `phasorTerm`)**.** $G_{\chi,N}(s) = \sum_{n<N} \chi(n)\, n^{-s}$*, the partial sums of the carrier-riding phasors.*

Theorem 5.6 (Accumulation on $\mathrm{Re}\, s > 1$; `fiber_accumulates_to_L`, `finiteCarrier_tendsto_LFunction`)**.** *For every Dirichlet character* χ *and* $\mathrm{Re}\, s > 1$*,* $G_{\chi,N}(s) \to L(s,\chi)$ *as* $N \to \infty$*.*

6 The three-channel three-dimensional phasor

The planar phasor $\chi(n)\, n^{-\sigma} \exp(-(y\log n)i)$ of Section 5 is a 2D arrow whose length is $\|\chi(n)\|\, n^{-\sigma}$; on the *neutral* channel $\chi(n) = 0$ it collapses to 0 and the magnitude $n^{-\sigma}$ is lost. The carrier, by contrast, places *every* slot n (Remark 3.2), neutral or not. The reconciliation lifts each term to a genuine vector in $\mathbb{R}^3 = (\text{spin plane } \mathbb{R}^2) \times (\text{mass axis } \mathbb{R})$, modeled as $\mathbb{C} \times \mathbb{R}$.

Definition 6.1 (3D phasor and its three channels; `Phasor3D.phasor3D`, `channel`)**.**

$$\mathrm{phasor3D}(\chi,\sigma,y,n) = \Big(\underbrace{\chi(n)\, n^{-\sigma} \exp(-(y\log n)i)}_{\textit{spin plane}},\; \underbrace{(1 - \|\chi(n)\|)\, n^{-\sigma}}_{\textit{mass axis}}\Big).$$

The charged channels $\chi(n) = \pm 1$ *place the rotating arrow* $\pm n^{-\sigma} \exp(-(y\log n)i)$ *in the spin plane with zero mass; the neutral channel* $\chi(n) = 0$ *places the real amplitude* $n^{-\sigma}$ *on the mass axis with no spin.*

Theorem 6.2 (The spin plane carries L unchanged; `phasor3D_plane_tsum`)**.** *The spin-plane (complex) coordinate of* phasor3D *is exactly the planar phasor of Section 5, so its accumulation is the same Dirichlet L-series. The 3D lift changes nothing in the analytic representation — it only records what the plane discards.*

Theorem 6.3 (Magnitude conservation across all channels; `phasor3D_mag3`)**.** *For a quadratic character the full* $\mathbb{R}^3$ *Euclidean length of every term is exactly* $n^{-\sigma}$*, on* all three *channels including the neutral one — where the planar length* $\|\chi(n)\|\, n^{-\sigma}$ *vanished. The magnitude lost from the plane is precisely the mass stored on the axis.*

Theorem 6.4 (Standing mass vs. travelling spin; `phasor3D_neutral_no_spin`, `phasor3D_charged_spin`, `resonantHeight_eq_exp_mellinReadoutFreq`)**.** *The neutral term is independent of the frequency* y *(**`phasor3D_neutral_no_spin`**): a* standing *(resonant) mode, not a travelling one, storing* *positive amplitude $n^{-\sigma}$ on the mass axis (`phasor3D_neutral_mass_pos`) while contributing 0 to the spin-plane L-series (`phasor3D_neutral_plane_zero`). The charged terms rotate at the readout rate* $\log n$ *as y advances (`phasor3D_charged_spin`) — the Mellin/readout frequency, distinct from the geometric carrier spin $n \cdot (\pi/3)$. The standing mass resonates at this readout frequency* $\mathrm{mellinReadoutFreq}(n) = \log n$ *and height* $\mathrm{resonantHeight}(n) = n = e^{\log n}$ *(`resonantHeight_eq_exp_mellinReadoutFreq`), matching the carrier height law $z_n = n = e^{y_n}$ of Remark 3.2.*

Remark 6.5 (Three channels, one decomposition; `phasor3D_three_channel_form`). *Every term lies in exactly one channel — positive ($\chi(n) = +1$), negative ($\chi(n) = -1$), neutral ($\chi(n) = 0$) — and is determined by it (`phasor3D_three_channel_form`). The signed pair (positive/negative) is the spinning content the L-series sees; the neutral channel is the mass the planar model omits. This positive/negative/neutral split is the channel structure built into the harmonic pencil of Section 15.*

7 Strip extension: exact accumulation into $\mathrm{Re}\, s > 0$

The fiber does not merely approximate the analytic object near the critical line. After Abel summation it has an exact limit throughout the half-plane $\mathrm{Re}\, s > 0$: for non-principal characters that limit is $L(s, \chi)$ itself, and for the principal channel it is the exact eta-regularized zeta value. Thus the critical-line readout is reached from a proved strip identity, not from a finite truncation.

Theorem 7.1 (Dirichlet strip extension; `dirichlet_strip_tendsto_LFunction`). *For a Dirichlet character $\chi \neq 1$ modulo q and $\mathrm{Re}\, s > 0$,*

$$\sum_{n<N} \chi(n)\, n^{-s} \xrightarrow[N\to\infty]{} L(s, \chi).$$

The convergence target is exactly Mathlib's Dirichlet L-function, not an auxiliary regularization.

Theorem 7.2 (Eta strip extension; `eta_strip_tendsto`). *For $\mathrm{Re}\, s > 0$ and $s \neq 1$,*

$$\sum_{n<N} (-1)^{n+1}\, n^{-s} \xrightarrow[N\to\infty]{} \left(1 - 2^{1-s}\right)\zeta(s).$$

Proof sketch. Bucket cancellation gives a uniform bound on the character sums $\sum_{n<N} \chi(n)$; Abel summation by parts [6, 3] then yields convergence of $\sum \chi(n) n^{-s}$ on $\mathrm{Re}\, s > 0$, and the limit agrees with $L(s, \chi)$ on $\mathrm{Re}\, s > 1$ and hence everywhere on the connected strip by the identity theorem. For the alternating weight the factor $1 - 2^{1-s}$ is the even-term correction relating η and ζ. □

Theorem 7.3 (Continuous reach to the line; `continuous_model_zeta`). *For every $y \in \mathbb{R}$, at $s = \frac{1}{2} + iy$,*

$$\sum_{n<N} (-1)^{n+1}\, n^{-s} \longrightarrow \left(1 - 2^{1-s}\right)\zeta\left(\tfrac{1}{2} + iy\right),$$

and the limit is 0 if and only if $\zeta(\frac{1}{2} + iy) = 0$.

Proof sketch. Theorem 7.2 at $s = \frac{1}{2} + iy$ (using $\mathrm{Re}\, s = \frac{1}{2} > 0$ and $s \neq 1$). On the line the factor $1 - 2^{1-s}$ is nonzero (`correction_factor_ne_zero_on_critical_line`), so the product vanishes exactly when $\zeta(\frac{1}{2} + iy)$ does (`etaTrivial_eq_zero_iff`). □

7.1 Per-height convergence: the cancellation is in the fiber, not the carrier

The accumulation converges height-by-height, and its vanishing is a property of the *fiber*, never of the carrier. The two objects are distinct. The carrier $\mathrm{helixPt}(n) = \sqrt{n}\,\mathrm{wind}(n)$ has modulus exactly $\sqrt{n}$, independent of the ordinate y and the character χ (`carrier_norm_height_independent`), and never vanishes off the origin (`carrier_ne_zero`): it is a no-radial-drift source. The fiber paired against it is the unit-modulus spin (`fiber_spin_unimodular`) times the arithmetic datum $\chi(n)\,n^{-1/2}$; attaching it only *rotates* the carrier and cannot alter the $\sqrt{n}$ radial profile. Hence every vanishing of an accumulation is a fiber/pairing cancellation (orthogonality, Proposition 14.8), never a collapse of the carrier.

Theorem 7.4 (Per-height channels; `finiteCarrier_critical_tendsto`, `eta_critical_tendsto`, `perHeight_channel_nonprincipal`, `perHeight_channel_zeta`)**.** *For every ordinate $y \in \mathbb{R}$:*

(i) *for non-principal χ the raw weighted partial sums $\sum_{n<N} \chi(n)\, n^{-(1/2+iy)}$ converge to $L(\frac{1}{2} + iy, \chi)$ (`finiteCarrier_critical_tendsto`), so the convergent channel's nonvanishing factor is 1;*

(ii) *for the principal character (and ζ) the raw series diverges, but the alternating (eta) channel $\sum_{n<N}(-1)^{n+1}n^{-s}$ converges to $(1-2^{1-s})\zeta(s)$, whose correction factor is nonzero on the line (`eta_critical_tendsto`, `zeta_eta_factor_ne_zero`).*

In both cases the channel value vanishes exactly when the L-value does (`LFunction_critical_eq_zero_iff_pairing` `riemannZeta_critical_eq_zero_iff_eta_tendsto_zero`), the on-line limit $y \mapsto L(\frac{1}{2} + iy, \chi)$ is continuous (`continuous_LOnLine`), and it is the analytic limit of the finite carrier (`finiteCarrier_tendsto_LOnLine`).

This is the precise, defensible form of "every zero on the line is subject to the cancellation machine": it asserts nothing about *where* the zeros are, only that each on-line zero is exactly a fiber cancellation of the convergent channel.

7.2 A fully convergent worked fibre for ζ

For the Riemann zeta channel the convergence above is realized as an explicit, quantitatively controlled Cauchy sequence. Place the fibre at the critical-line point $\mathrm{sFiber}(T)$, which has real part $\frac{1}{2}$ by construction (`sFiber_re`).

Theorem 7.5 (The η-fibre converges on the line; `Pi3Fiber.PartialFiber_tendsto_FullFiber`, `PartialFiber_cauchySeq`, `FullFiber_eq_zero_iff_zeta`)**.** *The alternating partial sums $\mathrm{PartialFiber}(N,T) = \sum_{n<N}(-1)^n\,(n+1)^{-\mathrm{sFiber}(T)}$ form a Cauchy sequence (`PartialFiber_cauchySeq`) converging to $\mathrm{FullFiber}(T) = \eta(\mathrm{sFiber}(T))$ (`PartialFiber_tendsto_FullFiber`, with vanishing tail `TailFiber_tendsto_zero`); and it vanishes* iff $\zeta(\mathrm{sFiber}(T)) = 0$ *(`FullFiber_eq_zero_iff_zeta`). The phasor form of the partial sum exhibits the spin* $\exp(-(HT\log(n+1))i)$ *on the amplitude* $(n+1)^{-1/2}$ *(`PartialFiber_eq_phasor`).*

Proof sketch. The summand differences are controlled by the consecutive-power bound $\|a^{-w} - (a+1)^{-w}\| \le \|w\|\,a^{-\operatorname{Re} w - 1}$ for $a \ge 1$, $\operatorname{Re} w > 0$ (`cpow_neg_consecutive_norm_le`), giving summability of the bracketed series (`pairedTerm_summable`) and the Cauchy property on $\operatorname{Re} s > 0$. The bracketed sum agrees with η for $\operatorname{Re} s > 1$ and extends by the identity theorem across the preconnected region $\{\operatorname{Re} s > 0\} \setminus \{1\}$ (`pairedEta_eq_etaTrivial`, `rightHalfPlane_sdiff_one_isPreconnected`, `sFiber_ne_one`). □

Remark 7.6 (Exact accumulation, not approximate evaluation)**.** *The critical-line behavior of L is classically reached by* approximate *means — the approximate functional equation and the Riemann–Siegel formula — which represent $L(\frac{1}{2}+it)$ by truncated sums with a controlled error term. The accumulation here is* exact*: the regularized carrier converges to L on the line (Theorems 7.4, 7.5) and vanishes* iff *L does (Theorem 9.5), an identity with no error term, machine-checked. Correspondingly the vanishing is an exact* event *— the sampled eigenwave is exactly ℓ^2-orthogonal to the arithmetic data (Proposition 14.8) — rather than an approximate minimum of $|L|$. The exactness is what lets the carrier's geometric and spectral structure transfer to L rigorously rather than heuristically; it is the foundation the rest of the development stands on. (The underlying convergence is the classical Abel/η summation on $\operatorname{Re} s > 0$; what is new is the exact, machine-checked geometric carrier built on it.)*

8 The geometric explicit formula

The logarithmic derivative of the fiber is the prime-power accumulation weighted by the winding.

Theorem 8.1 (Prime field; `explicit_formula_prime_field`, `dirichlet_logDeriv_eq_tsum`)**.** *For $\operatorname{Re} s > 1$,*

$$-\frac{L'(s,\chi)}{L(s,\chi)} \;=\; \sum_n \chi(n)\,\Lambda(n)\,n^{-s},$$

where Λ is the von Mangoldt function. The zeros of $L(\cdot,\chi)$ are the poles of the meromorphic continuation of this object.

Definition 8.2 (Residue harmonic; `Residue.residueHarmonic`)**.** *For $x > 0$ and $\gamma \in \mathbb{R}$, write $\rho = \frac{1}{2} + i\gamma$ and set* $\text{residueHarmonic}(x,\gamma) = -x^\rho/\rho$*.*

Theorem 8.3 (Zeros located as poles; `zeta_zero_located_as_pole`, `L_zero_located_as_pole`)**.** *Let $\rho = \frac{1}{2} + i\gamma$ be a zero of $L(\cdot,\chi)$ at which $L'(\rho,\chi) \neq 0$ (a simple zero). Then ρ is a simple pole of the explicit-formula kernel $-(L'/L)(s)\,x^s/s$, with*

$$\lim_{s\to\rho}(s-\rho)\left(-\tfrac{L'(s,\chi)}{L(s,\chi)}\cdot\tfrac{x^s}{s}\right) \;=\; \text{residueHarmonic}(x,\gamma) \;=\; -\frac{x^\rho}{\rho}.$$

The same statement holds verbatim for ζ.

Proof sketch. On the punctured neighborhood of ρ in $\{s \neq 1\}$, L is analytic with $L(\rho) = 0$ and $L'(\rho) \neq 0$, so $-(L'/L)$ has a simple pole at ρ with residue 1; multiplying by the analytic factor x^s/s scales the residue to x^ρ/ρ and the leading sign gives $-x^\rho/\rho$. □

9 Projection and reality on the critical line

The carrier is three-dimensional; the readout projects it to a one-dimensional coordinate. We first record that coordinate and the change of scale it carries (Section 9.1), then the reality of the completed object that the readout returns (Section 9.2).

9.1 The readout coordinate and its scaling

The carrier is built in its native gauge: the unit is $\pi/3$, and a full angular turn is six such units, $6\cdot(\pi/3) = 2\pi$ (`spin_phasor_period6`, Theorem 3.13). The readout that sends a three-dimensional harmonic value $t \in \mathbb{R}$ down to a one-dimensional coordinate composes a Cayley map [9] with a rescaling to the standard unit interval (`HarmonicProjection`):

$$\mathrm{toCircleAngle}(t) = 2\arctan t \in (-\pi, \pi), \qquad \mathrm{toLine}(\theta) = \frac{\theta + \pi}{2\pi} \in (0, 1),$$

and projection = toLine∘toCircleAngle. The second map carries the change of scale: it sends the full turn 2π to length 1, so the six $\pi/3$ buckets of the gauge become six equal sub-intervals of the readout. Under this rescaling the coordinate is rigid — the Cayley map is odd (`toCircleAngle_odd`), the normalization is affine hence midpoint-preserving (`toLine_midpoint`), and the principal-arc ends $\theta = \pm\pi$ map to the two ends of the interval (`toLine_neg_pi`, `toLine_pi`) — so the centre of the gauge is carried to the centre of the interval.

Theorem 9.1 (Coordinate centre; `HarmonicProjection.projection_midline`)**.** projection(0) *is the midpoint of the readout interval: the centre of the six gauge buckets, the angle $\theta = 0$, maps to the average of the two endpoint images (`toLine_midpoint_of_endpoints`).*

The inverse conversion is the same scaling read backwards: a readout coordinate u is the angle $\theta = 2\pi u - \pi$, the gauge reading $6u - 3$ in $\pi/3$ units, and the interval's centre is the gauge centre $\theta = 0$. This is a property of the readout coordinate and its scaling. The value the readout returns on the completed object is real — it lands on the real axis (Section 9.2).

9.2 Reality on the line

The supporting analytic facts are the conjugation (Schwarz reflection [9]) and functional symmetries of Λ. The $\frac{1}{2}$ in this subsection is the *reflection axis* $\mathrm{Re}\, s = \frac{1}{2}$ — the fixed line of $s \mapsto 1 - s$ on which $\overline{s} = 1 - s$ (Section 2) — *not* the scale-critical exponent of Section 3.3: the reality is read off the functional equation, not the area law.

Theorem 9.2 (Conjugation symmetry; `HelixCollapse.completedRiemannZeta_conj`)**.** *For all $s \in \mathbb{C}$, $\overline{\Lambda(\overline{s})} = \Lambda(s)$.*

Proof sketch. On $\mathrm{Re}\, s > 1$ this is the real Dirichlet coefficients of Λ ($\overline{\Lambda(\overline{s})} = \Lambda(s)$ termwise, using $\overline{\pi^{-s/2}} = \pi^{-\overline{s}/2}$ and $\overline{\Gamma(\overline{s}/2)} = \Gamma(s/2)$); the identity extends to $\mathbb{C}$ by analytic continuation of Λ_0 (the pole-removed completion) via the identity theorem. □

Theorem 9.3 (Reality on the line; `completedRiemannZeta_critical_line_im_zero`)**.** *For every $t \in \mathbb{R}$, $\mathrm{Im}\, \Lambda(\frac{1}{2} + it) = 0$.*

Proof sketch. On the line $\overline{\frac{1}{2} + it} = 1 - (\frac{1}{2} + it)$, so Theorem 9.2 together with $\Lambda(1 - s) = \Lambda(s)$ (`completedRiemannZeta_one_sub`) gives $\overline{\Lambda(\frac{1}{2} + it)} = \Lambda(\frac{1}{2} + it)$, i.e. a real value. □

Theorem 9.4 (Faithful projection for ζ; `faithful_projection_zeta`)**.** *For every $\gamma \in \mathbb{R}$,*

$$\mathrm{Im}\, \Lambda\big(\tfrac{1}{2} + i\gamma\big) = 0 \qquad \textit{and} \qquad F_\eta(\gamma) = 0 \iff \zeta\big(\tfrac{1}{2} + i\gamma\big) = 0,$$

where $F_\eta(\gamma) = (1 - 2^{1-s})\zeta(s)$ at $s = \frac{1}{2} + i\gamma$ (`EtaTrivial.Feta`).

Theorem 9.5 (Located crossings lie on the line; `crossing_is_zero_on_line`)**.** *If the eta-regularized limit of Theorem 7.3 vanishes at ordinate y, then $\zeta(\frac{1}{2} + iy) = 0$ and $\mathrm{Re}(\frac{1}{2} + iy) = \frac{1}{2}$.*

10 Faithfulness for every Dirichlet L-function and Tate completion

A single carrier serves all χ; the character weight $\chi(n)$ on each phasor is the only dial.

Theorem 10.1 (One carrier, all characters; `faithful_all_L`)**.** *For every Dirichlet character χ:*

(i) *for* $\operatorname{Re} s > 1$, $\sum_{n<N} \chi(n)\, n^{-s} \to L(s, \chi)$*; and*

(ii) *on the critical line* $\operatorname{Re} s = \frac{1}{2}$*, the eta-twisted closed carrier* `etaTwistClosed`(χ, s) *vanishes if and only if* $L(s, \chi) = 0$ *(*`etaTwistClosed_eq_zero_iff_critical`*).*

Definition 10.2 (Completed carrier; `Tate.completedCarrier`)**.** *For a Dirichlet character χ, set* $\Lambda^{c}_{\chi}(y) = \Lambda(\chi, \frac{1}{2} + iy)$ *using the archimedean* Γ*-completion* $\Lambda(\chi, s) = (\textit{gamma factor}) \cdot L(s, \chi)$*.*

Theorem 10.3 (Completion and functional equation; `faithful_all_L_completed`, `completedCarrier_eq_zero_i` `completed_functional_equation`)**.** *For every* primitive χ *modulo q:*

(i) *the completion is zero-free off the zeros of L on the line:* $\Lambda^{c}_{\chi}(y) = 0 \iff L(\frac{1}{2} + iy, \chi) = 0$ *(the* Γ*-factor never vanishes for* $\operatorname{Re} s > 0$*); and*

(ii) *Tate's functional equation [12] holds:* $\Lambda(\chi, 1 - s) = q^{\,s-1/2}\, W(\chi)\, \Lambda(\chi^{-1}, s)$*, where* $W(\chi) =$ `rootNumber`(χ)*.*

11 The double-ended carrier and conjugacy

The carrier is chiral: the right strand winds with spin $e^{-iy \log n}$, and its *conjugate* — the anti-helix, the left strand — winds with $e^{+iy \log n}$. Each strand is chiral; the symmetry belongs to the *pair*, and the origin is the midpoint of the double structure (Section 12).

Theorem 11.1 (Conjugate chiralities; `both_helices_conjugate`)**.** *For every $N \in \mathbb{N}$ and $y \in \mathbb{R}$,*

$$\sum_{n<N} (-1)^{\,n+1}\, n^{-(1/2-iy)} \;=\; \overline{\sum_{n<N} (-1)^{\,n+1}\, n^{-(1/2+iy)}}\,.$$

Proof sketch. Termwise: $\overline{(-1)^{n+1}} = (-1)^{n+1}$ and $\overline{n^{-(1/2+iy)}} = n^{-(1/2-iy)}$ for the positive real base n (so $\arg n \neq \pi$), by `Complex.cpow_conj` and $\overline{n} = n$. □

Corollary 11.2 (Equal modulus, identical crossing heights)**.** *The two strands have equal modulus at every N, so their limits vanish at the same ordinates y.*

Theorem 11.3 (Completed sides agree; `both_sides_cross_together`)**.** *For every* $y \in \mathbb{R}$, $\Lambda(\frac{1}{2} - iy) = \Lambda(\frac{1}{2} + iy)$*.*

Proof sketch. $\frac{1}{2} - iy = 1 - (\frac{1}{2} + iy)$, so this is $\Lambda(1 - s) = \Lambda(s)$ (`completedRiemannZeta_one_sub`) on the line. □

12 Criticality as midpointness, and the self-dual readout

Two structural facts sharpen the ontology. *Criticality is a midpoint statement with no number in it* — the familiar $\frac{1}{2}$ is a chart constant; and *the readout is self-dual* — the Fourier-conjugate variable of the readout line is the carrier's own height axis, so the ambient spectroscopy of Section 19.7 inverts the projection rather than adding a dimension.

12.1 UNIT/2: the chart-free form of criticality

Theorem 12.1 (Midpoint charts; `real_axis_is_conjugation_midpoint`, `centered_chart`, `riemann_chart`). *The fixed locus of complex conjugation is exactly the real axis* ($\overline{z} = z \iff \operatorname{Im} z = 0$) *— a criticality statement containing no number. For any unit c, reality of the centered variable* $t = -i(\rho - c/2)$ *is exactly* $\operatorname{Re}\rho = c/2$*; the case* $c = 1$ *is Riemann's own chart:* "the roots of Ξ are real" and "the zeros have Re $= \frac{1}{2}$" are one statement in two charts*, and the* $\frac{1}{2}$ *is the translation constant between them, not content.*

Theorem 12.2 (UNIT/2 on the line, the circle, and in every base; `affine_reflection_fixed_iff`, `conj_fixed_iff`, `eisenstein_conj_axis`, `criticality_is_half_unit`). *The fixed point of the reflection* $x \mapsto c - x$ *is* $c/2$*, for every unit c. On the circle, the anti-involution* $z \mapsto e^{i\Delta}\overline{z}$ *fixes exactly the half-unit direction* $\Delta/2$*; for the carrier's cell* $\Delta = \pi/3$ *the conjugation axis is* $\pi/6$ *— the hexagon's edge-midpoint line, the same UNIT/2 formula in the carrier's own geometry. And in base-b counting* ($b > 1$) *the critical abscissa reads* $\sigma \log b = \log b/2$*: the log-7 system's midpoint is* $\log\sqrt{7}$*, and the base-free content is that the critical amplitude is the geometric mean,* $n^{-1/2} = \sqrt{1 \cdot n^{-1}}$*, of the dual pair.*

Across every chart the invariant is the same: *midpointness of an involution.* The reflection $s \mapsto 1 - s$ fixes exactly $\operatorname{Re} s = \frac{1}{2}$ (Theorem 17.5); the de Branges variable is real exactly there (Corollary 22.1); over $\mathbb{F}_q$ the balance forces $\|q^s\|^2 = q$, i.e. the Frobenius midpoint $q^{1/2}$ (`midpoint_forcing`). The coordinate is gauge; the involution is the content.

12.2 The self-dual Mellin pairing: the spectral axis is the height axis

Theorem 12.3 (Bohr orthogonality and site recovery; `char_mean_zero`, `char_mean_one`, `readout_dual_recovers_sites`). *For a finite bank with arrows* a_n *at distinct log-heights* h_n*, the Bohr mean of the readout against a test frequency recovers the arrow at that height:*

$$\frac{1}{T}\int_0^T \Big(\sum_n a_n\, e^{-ih_n y}\Big) e^{i\omega y}\, dy \xrightarrow[T\to\infty]{} \sum_{n\,:\,h_n=\omega} a_n,$$

and 0 *when no site sits at* ω*.*

The conjugate variable ω of the readout line is therefore *not a new dimension — it is the carrier's log-height axis*: the measured spectral lines at $\omega = \ln p^k$ (Section 19.7) are the arrows at the prime-power sites, recovered by inversion. Zeros are events in y; primes are lines in ω; the explicit formula (Section 8) is this one pairing read in both directions. The remaining coordinate of $s = \sigma + iy$ is frozen at UNIT/2 by scale balance (`scaleBalanced_iff`, Theorem 3.11) — which is why the strip/Abel machinery of Section 7 is a device of the one-dimensional projection only, with no three-dimensional counterpart.

Theorem 12.4 (The prime clocks are mutually incommensurable; `prime_clocks_incommensurable`). *For distinct primes* $p \neq q$*, the ratio* $\log p/\log q$ *is irrational. In the base-q system prime q ticks at unit rate and every other prime at an irrational rate: unique factorization, read as spectral non-collision — no two ambient lines can ever coincide.*

13 The hinge: turning point, weld ray, and the two-strand kernel

The origin of the double-ended carrier — the *hinge*, the midpoint of the conjugation involution (Sections 11, 12) — has a proved local structure: it is a turning point of the collapse wave, its phase ray reads the root number, its jet parity is the weld sign, and for degree two it exists only as the meeting point of two windowed strands. All statements in this section are kernel-checked (`HingeKernel`, `AntihelixWindow`, `BSDLadder`); their measured faces are reported in Sections 19.11 and 19.12.

Theorem 13.1 (The hinge is a turning point; `collapseWave_even`, `hinge_turning_point'`). *Let $Z(t)$ be the collapse wave — the real value of $\Lambda(\frac{1}{2}+it)$ on the line (`collapseWave`; reality is Theorem 9.3). One functional-equation line gives evenness, $Z(-t)=Z(t)$ (`completed_zeta_line_even`, `collapseWave_even`), and hence $Z'(0)=0$ — stated honestly as `HasDerivAt`: the wave is differentiable at the hinge (`collapseWave_differentiableAt`), and its genuine derivative there vanishes (`hinge_turning_point'`). The oscillator of Section 19.11 therefore starts at an extremum, not a crossing: the first vanishing sits a quarter cycle out (measured 0.499–$0.500\,\pi$), every later cell a half cycle.*

Theorem 13.2 (The weld pins the line phase to half the root number; `weld_pins_half_phase`, `weld_minus_one_forces_zero`). *A value satisfying the weld anti-involution $z=e^{i\Delta}\,\overline{z}$ — as $\Lambda(\frac{1}{2}+it,\chi)$ does with $\Delta=\arg\varepsilon$ at* every *t, by the functional equation plus conjugation — lies on the ray $\Delta/2$: this is `conj_fixed_iff` (Theorem 12.1ff.) applied at the weld, UNIT/2 on the circle. The measurable face is $\arg\Lambda(\frac{1}{2}+it,\chi)\equiv\arg(\varepsilon)/2 \pmod{\pi}$: the root number is read from the line phase (Section 19.12). And if the weld phase is -1, a real value fixed by $z\mapsto-\overline{z}$ must vanish (`weld_minus_one_forces_zero`): the structural kernel of the forced central zero at $\varepsilon=-1$.*

Theorem 13.3 (Jet parity: the hinge dimension has the weld's parity; `hingeDim_parity`). *For a wave with weld sign ε ($f(-t)=\varepsilon f(t)$, $\varepsilon=\pm1$), all jets of the opposite parity die at the hinge (`even_odd_jets_dead`, `odd_even_jets_dead`), so a live jet has the weld's parity (`even_live_jet_is_even`, `odd_live_jet_is_odd`). Writing $d(0)=\mathrm{hingeDim}$ for the first live jet (`hingeDim`, `hingeDim_mem`; everything below it vanishes, `dead_layers_below`),*

$$(-1)^{d(0)}=\varepsilon \qquad (\textit{hingeDim\_parity}),$$

and $\varepsilon=-1$ forces $d(0)\geq1$ (`hingeDim_pos_of_odd`) — the parity conjecture's analytic shadow, as unconditional single-function kernels. For ζ itself ($\varepsilon=+1$) every odd jet is dead at the hinge (`collapseWave_odd_jets_dead`).

Theorem 13.4 (The two-strand kernel and its window; `AntihelixWindow`). *The upper incomplete Gamma $\Gamma(s,x)=\int_x^\infty e^{-t}t^{s-1}\,dt$ (`upperGamma`) is a growth window in the sense of the absorption law: fully open at the cut $x=0$ (`upperGamma_zero`), nonnegative and monotone dying (`upperGamma_nonneg`, `upperGamma_antitone`), and vanishing in the limit (`upperGamma_tendsto_zero`); the complete $\Gamma(s)$ splits exactly into the two path-halves at any cut (`gamma_splits_at_cut`) — the geodesic between the two endings, cut at the midpoint. For the two-strand term $r^s\,\Gamma(s,x)+\varepsilon\,r^{2-s}\,\Gamma(2-s,x)$ (`strandPair`): the hinge-normalized strand weights satisfy $r^{s-1}\cdot r^{1-s}=1$ identically (`strand_weights_det_one`) — the Frobenius determinant-one conjugate block realized in the s-direction; the strand swap $s\mapsto2-s$ fixes exactly the hinge $s=1$ (`strand_swap_fixed_iff`); and at the hinge the two strands of* every *term are equal, so $\varepsilon=-1$ annihilates each phasor individually (`weld_kills_each_phasor`) — the forced central zero is term-local, not a conspiracy of the sum — while $\varepsilon=+1$ doubles each to $2re^{-x}$ (`weld_doubles_each_phasor`, via $\Gamma(1,x)=e^{-x}$, `upperGamma_one`). In the log-cusp coordinate the Fricke involution reads $u\mapsto-\log N-u$ and fixes exactly $u=-(\log N)/2$ (`fricke_midpoint_is_half_conductor_unit`): UNIT/2 with the conductor as the unit — the hinge is the midpoint between the two endings.*

Theorem 13.5 (The reference ladder; `Gr`, `Gr_nonneg`, `Gr_one_eq`)**.** *The ladder $G_0(x) = e^{-x}$, $G_{r+1}(x) = \int_x^\infty G_r(t)\,dt/t$ — whose sums $2\sum(a_n/n)\,G_r(2\pi n/\sqrt{N})$ are the leading data $L^{(r)}(1)/r!$ — is nonnegative rung by rung (`Gr_nonneg`): the reference series is a dissipation profile. Its first rung is the reverb kernel, $G_1 = \Gamma(0,\cdot) = E_1$ (`Gr_one_eq`): the BSD rate and the residue machinery of Section 19.9 share one kernel.*

14 The Frobenius similitude and the determinant identity

The map $n \mapsto pn$ — and more generally $n \mapsto mn$ — is the *Frobenius self-similarity* of the carrier. We define it precisely, as a similitude of the carrier line.

Definition 14.1 (Frobenius similitude; `frobeniusMultiplier`, `frobeniusSimilitude`)**.** *For an integer m the* Frobenius multiplier *is $\mu_\theta(m) = \sqrt{m}\,\mathrm{wind}_\theta(m) \in \mathbb{C}$, and the* Frobenius similitude $\mathrm{Sim}_\theta(m)$ *is the $\mathbb{C}$-linear self-map $w \mapsto \mu_\theta(m)\,w$ of the carrier line $\mathbb{C}$.*

Theorem 14.2 (Prime multiplication acts by similitude; `helixPt_mul`, `helixPt_eq_similitude`, `norm_frobeniusMultiplier`, `frobeniusSimilitude_mul`)**.** *For $m, n \neq 0$, multiplication $n \mapsto mn$ acts on the carrier point* $\mathrm{helixPt}_\theta(n) = \sqrt{n}\,\mathrm{wind}_\theta(n)$ *by the similitude*

$$\mathrm{helixPt}_\theta(mn) \;=\; \mathrm{Sim}_\theta(m)\big(\mathrm{helixPt}_\theta(n)\big) \;=\; \underbrace{\sqrt{m}}_{radial}\;\underbrace{\mathrm{wind}_\theta(m)}_{rotation}\;\mathrm{helixPt}_\theta(n).$$

Its multiplier has modulus $\|\mu_\theta(m)\| = \sqrt{m}$ — the area-law factor of Theorem 3.9, the winding rotation being unit-modulus — so the map scales every length by $\sqrt{m}$. Moreover the similitudes compose, $\mathrm{Sim}_\theta(mn) = \mathrm{Sim}_\theta(m) \circ \mathrm{Sim}_\theta(n)$*: the carrier is* self-similar *under the whole multiplicative monoid generated by the primes (the Frobenius maps).*

Under the bridge (Theorem 4.4), with prime-angle assignment $\theta(p) = -y \log p$ the rotation eigenphase at ordinate y is the spin $z = \mathrm{spin}(y, n) = e^{-iy\log n} \in \mathbb{T}$ (`frobeniusMultiplier_bridge`). The two chiralities (Theorem 11.1) carry z and $\overline{z}$.

Theorem 14.3 (Conjugate-eigenstate determinant; `frobenius_conjugate_det_one`)**.** *For every $y \in \mathbb{R}$ and $n \in \mathbb{N}$, with $z = \mathrm{spin}(y, n) = e^{-iy\log n}$,*

$$\det\begin{pmatrix} z & 0 \\ 0 & \overline{z} \end{pmatrix} \;=\; z\,\overline{z} \;=\; |z|^2 \;=\; 1.$$

Proof. The 2×2 determinant is $z \cdot \overline{z} - 0 \cdot 0 = z\,\overline{z}$. Since $\overline{z} = \overline{e^{-iy\log n}} = e^{\overline{-iy\log n}} = e^{+iy\log n}$ and $-iy\log n + \overline{-iy\log n} = 0$, we have $z\,\overline{z} = e^{-iy\log n}\,e^{+iy\log n} = e^0 = 1$. ☐

Theorem 14.4 (Unitary transverse block; `frobeniusBlock_det_one`, `frobeniusBlock_unitary`)**.** *The transverse block* $\mathrm{diag}(z, \overline{z})$ *assembled from the two chiral eigenphases at a conjugate crossing is* unitary, *$M^{\mathsf{H}}M = I$, with determinant $z\,\overline{z} = |z|^2 = 1$ — the orientation- and volume-preserving chiral invariant of the transverse phase dynamics. The radial scaling $\sqrt{m}$ of the similitude (Theorem 14.2) is carried separately and is not part of this block.*

14.1 The self-adjoint eigenstate

The determinant identity above is stated for the bare eigenphases $z, \overline{z} \in \mathbb{T}$. We realize them as the values of a genuine eigenstate of a self-adjoint generator. This is the operator *design*: it is real because its inputs are real, and it stands independently of any zero. Reading the nontrivial zeros as the spectrum of a self-adjoint operator is the *Hilbert–Pólya* program — supported by the random-matrix law of the zero statistics [13] and pursued through the Berry–Keating $H = xp$ quantization [14] and Connes's noncommutative-geometry trace formula [15]; the operator design here is unconditional and per-height, and the step from it to that canonical operator is exactly the named open link (Remark 14.11).

Definition 14.5 (Spectral wave; `spectralWave`)**.** *For $\gamma \in \mathbb{R}$, $\psi_\gamma(t) = e^{i\gamma t}$ ($t \in \mathbb{R}$).*

Theorem 14.6 (Unit-modulus eigenstate of $D = -i\,d/dt$; `spectralWave_eigen`, `spectralWave_norm`)**.** *For all $t \in \mathbb{R}$, $-i\,\psi'_\gamma(t) = \gamma\,\psi_\gamma(t)$ and $\|\psi_\gamma(t)\| = 1$. Thus ψ_γ is a unit-modulus eigenstate of the generator $D = -i\,d/dt$ with the real eigenvalue γ.*

The reality is structural, not an accident of ψ_γ. Multiplication by the real height, $\mathrm{vonNeumannOp}(\gamma) = \gamma \cdot \mathrm{id}$ on the fiber $\mathbb{C}$, is symmetric with eigenvalue γ (`vonNeumannOp_isSymmetric`, `vonNeumannOp_hasEigenvalue`); on the finitely-supported sequence space the real-diagonal operator $\mathrm{diagOp}(d)$ is symmetric (`diagOp_symmetric`), with the basis vectors as explicit eigenvectors carrying the real eigenvalues d_i (`diagOp_eigenvector`), and *every* eigenvalue is real (`diagOp_spectrum_real`) by von Neumann reality — a symmetric operator on a complex inner-product space has real eigenvalues (`symmetric_eigenvalue_real`). These are collected in `unconditional_frobenius_eigenstate` and `frobenius_spectralWave_eigenstate`. The eigenwave is a one-parameter unitary group, $\psi_\gamma(s+t) = \psi_\gamma(s)\,\psi_\gamma(t)$ with $\psi_\gamma(0) = 1$ (`spectralWave_add`, `spectralWave_zero`): the unitary flow generated by the real spectral parameter γ (Stone form).

Proposition 14.7 (The chiralities are eigenstate values; `spin_eq_spectralWave`, `conj_spin_eq_spectralWave`)**.** *For every $y \in \mathbb{R}$ and $n \in \mathbb{N}$, $\mathrm{spin}(y, n) = \psi_y(-\log n)$ and $\overline{\mathrm{spin}(y, n)} = \psi_y(\log n)$.*

The useful spectral content is not the mere existence of the eigenwave ψ_y — every real y gives one — but its *orthogonality* to the arithmetic data at a vanishing.

Proposition 14.8 (The accumulation is an eigenstate–data inner product; `fiberCarrierFinite_eq_inner`, `fiberCarrierFinite_eq_zero_iff_orthogonal`)**.** *Split the critical-line phasor $\chi(n)\,n^{-(1/2+iy)}$ into the arithmetic datum $\chi(n)\,n^{-1/2}$ (the finitely-supported vector **dataVec**(χ, N)) and the eigenwave sample $\overline{\mathrm{spin}(y, n)} = \psi_y(\log n)$ (the vector **waveVec**(y, N)). With the ℓ^2 inner product $\langle f, g\rangle = \sum_n \overline{f(n)}\,g(n)$ on the finitely-supported sequence space, the finite accumulation is their pairing,*

$$\sum_{n<N} \chi(n)\,n^{-(1/2+iy)} \;=\; \big\langle\, \textbf{waveVec}(y, N),\;\; \textbf{dataVec}(\chi, N)\,\big\rangle,$$

so it vanishes iff *the sampled eigenwave is ℓ^2-orthogonal to the arithmetic data (**fiberCarrierFinite_eq_zero_iff_** A vanishing point is exactly a real-frequency orthogonality event, exhibited at each N (and the pairing is the partial phasor sum $G_{\chi,N}(\frac{1}{2} + iy)$, which accumulates to the carrier value $L(\frac{1}{2} + iy, \chi)$; **inner_eq_finiteCarrier**). The Gram form of any finite family of such vectors is positive-semidefinite (**carrier_gram_nonneg**), the positive-kernel background for the de Branges evaluation of Section 22.3.*

Theorem 14.9 (Determinant on the eigenstate values; `frobenius_spectralWave_det_one`)**.** *For every $y \in \mathbb{R}$ and $n \in \mathbb{N}$, $\det \mathrm{diag}\big(\psi_y(-\log n),\, \psi_y(\log n)\big) = 1$.*

Theorem 14.10 (Frobenius unimodularity of the produced eigenstate; `frobenius_eigenstate_det_one`). *Fix $\gamma \in \mathbb{R}$, $n \in \mathbb{N}$, and $\psi : \mathbb{R} \to \mathbb{C}$, and* assume *the vanishing* $\Rightarrow$ *eigenstate link $\psi = \psi_\gamma$. Then ψ is a unit-modulus eigenstate of $D = -i\, d/dt$ with eigenvalue γ, and* $\det \operatorname{diag}\big(\psi(-\log n),\, \psi(\log n)\big) = 1$.

Remark 14.11 (The open link). *We exhibit the eigenstate ψ_γ for every real γ unconditionally (Theorem 14.6). What is* not *proved is that a carrier vanishing at height γ* produces *this eigenstate; that hypothesis ($\psi = \psi_\gamma$) is the one input of Theorem 14.10, left open. We do not derive that the vanishings are the spectrum of a canonical operator.*

15 The harmonic pencil and the focal eigenheight

The positive/negative channels of the 3D phasor (Section 6) assemble into a 2×2 determinantal pencil whose rank-drop locates the zeros — built in the geometric height $Z = e^y > 0$ rather than in $\log n$ (the *no-log-trap* design). The readout dictionary is $t = \log Z$, and the critical readout $\mathrm{criticalReadout}(Z) = \frac{1}{2} + i \log Z$ has real part $\frac{1}{2}$ for every real Z (`criticalReadout_re`).

15.1 The scalar channel closure

Definition 15.1 (Signed and unsigned harmonic channels; `HarmonicCell.Achan`, `Bchan`, `Focal.Pchan`, `Mchan`). *With amplitude $a_n(t) = (\pi/3)\, n^{-1/2} \exp(-it \log n)$, the positive and negative scalar channels are $P(t) = \sum_{\chi(n)=+1} a_n(t)$ and $M(t) = \sum_{\chi(n)=-1} a_n(t)$. The* signed *harmonic mode is $B_\chi(Z) = (\pi/3)\, L(\frac{1}{2} + i \log Z, \chi)$ — the Abel limit of the channel difference $P - M$. The factor $\pi/3$ is the exact six-cell unit of the carrier (`Ucell`); it is nonzero (`Ucell_ne_zero`), so multiplying by it preserves the zero set exactly. The* unsigned *mode is $A_\chi(Z) = L(\frac{3}{2} + i \log Z, \chi)$, read at the absolute-convergence abscissa $\sigma = \frac{3}{2}$ where it never vanishes (`Achan_ne_zero`).*

Theorem 15.2 (Channel closure $\iff$ L-zero; `Focal.channel_diff_tendsto`, `channel_closure_iff_L_zero`). *The scalar channel difference is the readout phasor partial sum, with Abel limit $(\pi/3)\, \Phi_\chi(Z) = (\pi/3)\, L(\frac{1}{2} + i \log Z, \chi)$ (`channel_diff_tendsto`). Hence the channel closes — the difference $P(\log Z) - M(\log Z)$ tends to 0 —* iff *$L(\frac{1}{2} + i \log Z, \chi) = 0$ (`channel_closure_iff_L_zero`). The $\pi/3$ scaling is what makes this a literal equality of cancellation events: the harmonic channel is a fixed nonzero multiple of the exact L-fiber, not a numerically tuned proxy.*

15.2 The determinantal pencil

Theorem 15.3 (Harmonic pencil determinant; `HarmonicCell.harmonicPencil_det`, `harmonic_pencil_det_zero`
For the diagonal harmonic pencil

$$H_\chi = \begin{pmatrix} A_\chi & B_\chi \\ \mu A_\chi & \lambda B_\chi \end{pmatrix}, \qquad \det H_\chi = (\lambda - \mu)\, A_\chi B_\chi.$$

Under admissibility ($A_\chi \neq 0$, $\lambda \neq \mu$) the rank-drop $\det H_\chi = 0$ is exactly the scalar L-zero closure $L(\frac{1}{2} + i \log Z, \chi) = 0$.

Theorem 15.4 (Gram rank-drop is the same event; `gram_rank_drop_iff_det_cell_zero`, `gramH_rank_drop_iff_L_zero`, `helix_eigenheight_real`). *For any cell pencil L^{c} the Gram matrix $G^{\mathrm{c}} = (L^{\mathrm{c}})^{\mathsf{H}} L^{\mathrm{c}}$ is Hermitian positive semidefinite with the same kernel as L^{c}, and in the square case $\det G^{\mathrm{c}} = |\det L^{\mathrm{c}}|^2$; so the Gram rank-drop $\det G^{\mathrm{c}}_{\chi,h} = 0$ is again exactly the L-zero closure (`gramH_rank_drop_iff_L_zero`). The generalized eigenheights of such a positive pencil are real Rayleigh quotients $Z = \langle v, Av\rangle / \langle v, Bv\rangle$ (`helix_eigenheight_real`, `helix_eigenheight_rayleigh`), the no-log-trap height diagonal being $Z_n = n$, not $\log n$.*

15.3 The pencil as a family, and the finite crutch-free layer

The pencil is not a single matrix but a *family* of harmonic Gram interlaced operators $\{H_\chi(\mu, \lambda)\}$ over the admissible calibrations, and the rank-drop event is calibration-independent: all admissible members drop simultaneously, exactly when the signed channel closes (`gram_rank_drop_calibration_independent`; the event is governed solely by B_χ, independent of the unsigned mode — `gramH_rank_drop_iff_Bchan_zero`). Each member's Gram is Hermitian positive semidefinite (`gramH_posSemidef`) and is *never the zero matrix*: the rank drops by exactly one — a single cancellation, never a simultaneous double one (`specGram_apply00`, `specGram_ne_zero`; Section 16).

The detector itself is channel-agnostic — pure matrix algebra on a supplied pair, with no L-function in the statement: for any $A \neq 0$ and $\lambda \neq \mu$, $\det H(A, B; \mu, \lambda) = 0 \iff B = 0$, and likewise for the Gram (`harmonic_pencil_det_zero_iff_channel_zero`, `harmonicGram_rank_drop_iff_channel_zero`, `harmonicGram_rank_drop_calibration_independent`). Two instantiations are formalized. The *analytic* channels are $A_\chi = L(\frac{3}{2} + i \log Z, \chi)$ and $B_\chi = (\pi/3)L(\frac{1}{2} + i \log Z, \chi)$, with admissibility unconditional at the absolute-convergence abscissa (`Achan_ne_zero`). The *finite* channels are the live phasor-bank sums

$$A_\chi^{\text{fin}}(a, N) = \sum_{n=1}^{N} \|\chi(n)\| \, a_n, \qquad B_\chi^{\text{fin}}(a, \varphi, N) = \sum_{n=1}^{N} \chi(n) \, a_n \, e^{i\varphi_n},$$

with *arbitrary* magnitude window a and phase profile φ; here admissibility is discharged by *positivity* — $\chi(1) = 1$ and $a_1 > 0$ force $A^{\text{fin}} > 0$ — with no analytic input (`finiteA_pos`, `finiteA_ne_zero`), and the family theorems specialize verbatim (`finite_gramH_posSemidef`, `finite_gramH_rank_drop_iff_channel_zero`, `finite_gram_rank_drop_calibration_independent`). A separate module re-houses the pencil test entirely on the finite side — L appears only in the final-verification bridge: the finite channels converge to the analytic ones (`AchanFin_tendsto`, `BchanFin_tendsto`), the finite signed channel closes in the growth limit iff $L(\frac{1}{2} + i \log Z, \chi) = 0$ (`BchanFin_closure_iff_L_zero`), and the asymptotic finite-pencil rank-drop is equivalent to the L-zero (`finite_pencil_rank_drop_iff_L_zero`).

15.4 Admissibility of the source-height readout

Theorem 15.5 (Real source-height ⇒ critical line; `Admissible.admissible_iff_re_half`, `admissible_real_height_implies_critical_readout`, `nontrivial_zero_represented`)**.** *A complex point is the readout* criticalReadout(Z) *of a real source-height Z* iff *it lies on the critical line* Re $= \frac{1}{2}$ *(`admissible_iff_re_half`); the backward direction realizes it at $Z = e^{\operatorname{Im}\rho}$. Equivalently an off-line ρ admits* no *real source-height (`offline_not_admissible`): the height solving the readout is real iff* $\operatorname{Re}\rho = \frac{1}{2}$ *(`height_im`, its imaginary part being $-(\operatorname{Re}\rho - \frac{1}{2})$). Consequently every nontrivial* critical-line *zero is represented by an admissible real source-height Gram eigen-event $Z = e^t > 0$ (`nontrivial_zero_represented`).*

Remark 15.6 (Every zero has a source — the sourceless-zero question; `EveryZeroHasSource`, `everyZeroHasSource_iff_critical`, `zero_source_admissibility_of_everyZeroHasSource`)**.** *The readout reaches* exactly *the critical line for real Z (`criticalReadout_re`), so the 3D carrier's projection cannot land off-axis and cannot host an off-line zero; Theorem 15.5 represents every zero* presented on *the line. The hypothesis* every zero has a source *(`EveryZeroHasSource`) asserts that every nontrivial zero arises this way — from a real source height — and its source form is proved to be exactly the critical-line statement (`everyZeroHasSource_iff_critical`), hence GRH for χ; conditional on it,* every *nontrivial zero (with no on-line hypothesis) is a represented admissible eigen-event (`zero_source_admissibility_of_everyZeroHasSource`, `RH_of_everyZeroAdmissible`).*

The same hypothesis is the representability bridge `EveryZeroAdmissible`*; these two are the* formalized *faces of the one undecided question (Remark 17.7), whose completed-ζ form — and standing name — is* projection completeness *(*`projection_complete_iff_RH`*, Theorem 18.7). The de Branges* IsHB Λ *condition is the same RH content in the classical de Branges form (Remark 22.2), which this work connects to but neither formalizes nor relies on — open work in the de Branges program, not a conditional of this paper. The open question itself is posed, never assumed: RH fails only if a factor outside the 3D model produces an off-line zero, which the carrier provably cannot. As an unconditional complement, the perturbation (Rouché-type) stability of the focal zeros is fully proved: at a simple critical-line zero γ where the L-channel dominates the bridge error there is a unique nearby focal zero, with the explicit location bound $|T_d - \gamma| \le 2\varepsilon/a$ (*`real_simple_zero_location`*,* `real_simple_zero_exists`*,* `real_simple_zero_unique`*).*

16 The self-adjoint realization and the GRH equivalence

The spectral layer realizes the cancellation events as kernel events of a fixed operator and then isolates, exactly, the RH-strength statement. The 3D spectral-fiber readout converges to the analytic L-function (`spectral_fiber_is_Lfunction`); a zero is a kernel event of the configured spectral operator, and conversely (`spectral_zero_has_kernel`, `spectral_kernel_iff_zero`, `cancellation_equivalence`). The generator A_χ is the real-diagonal operator with eigen-heights $\log(n+1)$ — strictly increasing because *only one phasor is growing continuously at a time* (`logGen_strictMono`); this is the formal counterpart of the partial-absorption law of Section 19.1.

Theorem 16.1 (No double cancellation; `no_double_cancellation`, `cancellation_modes_subsingleton`, `cancellation_energy_not_split`)**.** *At a fixed spectral parameter s, at most one basis mode is a kernel vector of* specOp A_χ s*: two cancelling modes are the same mode. A cancellation event concentrates on a single site; the cancelled energy cannot be split between two distinct modes. Correspondingly the harmonic Gram matrix is never the zero matrix — its $(0,0)$ entry is $1 + \|\mu\|^2 \ge 1$ (*`specGram_apply00`*) — so the pencil family of Section 15.3 drops rank by exactly one per event.*

Theorem 16.2 (A realized cancellation is on the critical line; `nontrivial_Aχ_kernel_no_offline`)**.** *If s lies in the open critical strip and some basis mode is a kernel vector of* specOp A_χ s*, then* $\mathrm{Re}\, s = \frac{1}{2}$*. The mechanism is von Neumann reality: the kernel condition forces the spectral height* $\mathrm{specHeight}(s) = -i(s - \frac{1}{2})$ *to equal a real eigen-height, hence to be real.*

Theorem 16.3 (GRH from a self-adjoint realization; `GRH_from_nontrivial_Aχ_realization`, `GRH_from_spectral_exhaustion`)**.** *If every nontrivial cancellation of $L(\cdot, \chi)$ is realized as a kernel of the fixed self-adjoint generator A_χ, then every nontrivial zero lies on the critical line — GRH for χ. The implication is unconditional; the realization hypothesis is the isolated RH-strength input. The twin formulation through the sign-flip mechanism is equivalent: on the critical line the realization holds outright, and a real-generator cancellation forces* $\mathrm{Re}\, s = \frac{1}{2}$ *(*`nontrivial_spectral_cancellation_to_real_generator_of_re_half`*,* `realGeneratorCancellation_re_half`*).*

Remark 16.4 (The two-sided package, and where GRH sits)**.** *Theorem 15.5 realizes every* critical-line *zero as an admissible real-height eigen-event, unconditionally. Theorem 16.2 shows any* realized *nontrivial cancellation is on the critical line, unconditionally. The single unproved statement between them — that every nontrivial cancellation* is *realized — is the hypothesis that* every zero has a source*, whose source form is proved to be exactly the critical-line statement (*`everyZeroHasSource_iff_critical`*) and hence GRH for χ (Remark 17.7). GRH is thus the* *statement that the 1D limit has no zero without a kernel event of this explicit, elementary generator; which branch of the dichotomy holds is weighed, on the evidence, in Section 20.*

17 Admissibility: the carrier has no native off-axis support

The constructions above combine into a single intrinsic, on-axis statement about the carrier's *state space* — sharper than "a phasor plot that finds the zeros": it reframes the critical line as the carrier's admissible state space — the space of sources from which the zeros arise (Remark 17.7). A point of the vertical line $s = \sigma + iy$ rides the carrier as a stable, scale-normalized fiber only when its amplitude $n^{-\sigma}$ reciprocates the emergent area-law radius $r_n \sim \sqrt{n}$ to a finite positive scale. We call such a σ an *admissible (scale-balanced) state.*

Definition 17.1 (Scale-balanced admissible state; `ScaleBalanced`)**.** *For a carrier (p, r) and abscissa $\sigma \in \mathbb{R}$, σ is* scale-balanced *when the scale-balance product has a positive finite limit: $\exists L > 0,\ n^{-\sigma} r_n \to L$, where $r_n = \mathrm{carrierRadius}(p, r, n)$ is the emergent area-law radius (Definition 3.8, Theorem 3.9).*

Theorem 17.2 (The admissible exponent is forced, not chosen; `scaleBalanced_iff`, `scaleBalanced_setOf_eq`)**.** *For every carrier (p, r) with $r > 0$, σ is scale-balanced* iff *$\sigma = \frac{1}{2}$; equivalently $\{\sigma \in \mathbb{R} : \sigma \text{ scale-balanced}\} = \{\frac{1}{2}\}$. The critical abscissa is the* unique *exponent compatible with the arclength area law (Theorem 3.11), and the conclusion is gauge-independent: the gauge (p, r) fixes only the value of the limit, never which σ is admissible.*

Corollary 17.3 (No native off-axis support; `not_scaleBalanced_of_ne`)**.** *For $\sigma \neq \frac{1}{2}$ the fiber is* not *a scale-balanced state: its amplitude $n^{-\sigma}$ is not scale-normalized to the carrier (it decays faster than the radius grows, or outruns it). The carrier supports no admissible off-axis state.*

Remark 17.4 (Answering the "designed around $\frac{1}{2}$" objection)**.** *The exclusion of off-axis states is not built in: $\frac{1}{2}$ is* derived *from the arclength area law (Theorems 3.9, 3.11) as the unique exponent at which $n^{-\sigma}$ reciprocates $r_n \sim \sqrt{n}$. It is the $\frac{1}{2}$ of $\sqrt{n} = n^{1/2}$, not a posited abscissa.*

The functional equation supplies the global counterpart: the reflection $s \mapsto 1 - s$, which pairs the two chiral fibers, fixes exactly the critical line.

Theorem 17.5 (The reflection fixes exactly the critical line; `reflection_fixes_iff`)**.** *For $s \in \mathbb{C}$,* $\mathrm{Re}(1 - s) = \mathrm{Re}\, s \iff \mathrm{Re}\, s = \frac{1}{2}$. *Off-axis, $s \mapsto 1 - s$ sends an abscissa imbalance σ to the opposite imbalance $1 - \sigma$ — a dual* pair*, not a self-conjugate state — so the double-ended carrier closes only on the fixed locus* $\mathrm{Re}\, s = \frac{1}{2}$.

Theorem 17.6 (The hierarchy, assembled; `no_native_offaxis_support`)**.** *For every carrier (p, r) with $r > 0$, height y, and site n, the following hold simultaneously:*

(1) *the area law $r_n^2/n \to r(\pi/3)/\pi$ (so $r_n \sim \sqrt{n}$), from the arclength geometry;*

(2) *scale balance holds iff $\sigma = \frac{1}{2}$ (forced, not chosen);*

(3) *every $\sigma \neq \frac{1}{2}$ is inadmissible;*

(4) *at the balanced axis the chiral fibers $z, \overline{z}$ form a determinant-one* unitary *transverse block (Theorem 14.4); and*

(5) *$s \mapsto 1 - s$ fixes exactly* $\mathrm{Re}\, s = \frac{1}{2}$, *on which the de Branges variable is real (Corollary 22.1).*

The chain

$$r_n \sim \sqrt{n} \implies \sigma = \tfrac{1}{2} \implies \textit{unitary chiral block} \implies \det = 1 \implies s \leftrightarrow 1-s \textit{ closes on } \operatorname{Re} s = \tfrac{1}{2}$$

is the carrier's intrinsic logic: it reads as a geometry, not a graphing of $L(\frac{1}{2}+it)$.

Remark 17.7 (Projection completeness: the pivotal hypothesis)**.** *Theorem 17.6 establishes that the carrier's admissible state space — the image of its real-height projection $Z \mapsto \frac{1}{2} + i\tau_\chi(\log Z)$ — is the critical line: off-axis points are unsupported / inadmissible / not scale-normalized, so the 3D representation* cannot *host an off-line zero (`criticalReadout_re`). The 3D side is thereby closed: its readout values are exactly L on the strip, every event it produces is an on-line zero, and every on-line zero is one of its events. The pivotal hypothesis —* projection completeness*: every zero has a source (`EveryZeroHasSource`), whose source form is proved to be exactly the critical-line statement (`everyZeroHasSource_iff_critical`), hence GRH for χ, with the exhaustion unconditional given it (`zero_source_admissibility_of_everyZeroHasSource`) — is therefore a statement about the 1D analytic object alone. The dichotomy is asymmetric:* either *every zero of the 1D limit arises from a source event of the 3D structure,* or *the limit function admits a* sourceless *zero — a vanishing at an off-line point of the readout's chart, a coordinate the carrier does not possess (σ is a dial of the readout, not a place in the geometry), produced by no event. The Riemann Hypothesis is thus not something weaker that this work falls short of, but exactly the statement that the 1D readout adds no zeros of its own: the work closes the 3D side by theorems and localizes the unknown in the 1D limit. Section 18 formalizes this frame split; Section 20 weighs the empirical record on the remaining question. "Frobenius" here means precisely the prime-multiplication similitude of Definition 14.1.*

18 Weil duality, chirality, and the frame split

The double-ended helix admits a Weil–Deligne reading: the carrier is the geometric object, the vanishings are its "cohomology" — the eigenvalue spectrum of a Frobenius acting there. The dictionary — variety $\leftrightarrow$ carrier-plus-fiber; Frobenius eigenvalues on H^i $\leftrightarrow$ the vanishings; Poincaré duality $\alpha \leftrightarrow q^w/\alpha$ $\leftrightarrow$ the helix/antihelix pairing; Lefschetz trace formula $\leftrightarrow$ the medium spectroscopy; purity $|\alpha| = q^{w/2}$ $\leftrightarrow$ all vanishings on the conjugation axis — is carried at three explicitly separated tiers of honesty. *Literal*: at each prime, degree-two purity *is* Deligne's theorem (Ramanujan–Petersson via Weil II), and it appears in the model as reality of the clock angle — verified for all 2262 primes ≤ 20000 of Δ, whose helix is built literally from per-prime SU(2) rotors: $\lambda(p^k) = U_k(\cos\theta_p)$ exact to 2×10^{-15}, the Euler factor as the rotor's characteristic determinant, and Δ's zeros located from the rotor geometry alone [`tmp/gl2_helix.py`]. *Proven*: the duality package below. *Open, named*: purity of the global vanishing pairs — the step Deligne supplied geometrically, isolated below as one growth inequality. (Scope: the duality package is instantiated for the completed ζ and, at every conductor, for primitive Dirichlet L [`DirichletDuality`]: the pairing swaps the character — $\Lambda(1-\rho,\chi) = 0 \iff \Lambda(\rho,\chi^{-1}) = 0$, `vanishing_dual_pair_dirichlet`, exactly as the two-strand ontology says — and preserves the census dimension, $d_\chi(1-\rho) = d_{\chi^{-1}}(\rho)$ (`dual_dimension_symmetry_dirichlet`, via the jet-blindness workhorse `jets_mul_iff`), with the root-number nonvanishing carried as a visible hypothesis, `rootNumber_ne_zero_of_gaussSum`.)

Theorem 18.1 (Duality is a perfect det-one pairing on the vanishing set; `WeilDuality`)**.** *Λ vanishes at ρ iff at $1-\rho$ (`vanishing_dual_pair`) and iff at $\overline{\rho}$ (`vanishing_conj_pair` — the antihelix is the conjugate). The pairing preserves the local dimension: the k-th jet of Λ dies at $1-\rho$ iff it dies at ρ (`dual_dimension_symmetry`), so $d(\rho) = d(1-\rho)$ for the dimension function of the* *census (Section 19.12) — the jet towers die together, a perfect pairing on each class. And in any eigenvalue chart $c^{(\cdot)}$ ($c \neq 0$; the base is gauge), the hinge-normalized eigenvalues of a dual pair multiply to exactly one: $c^{\rho-1/2}\, c^{(1-\rho)-1/2} = 1$ (`dual_pair_det_one`) — the same determinant-one conjugate block as the similitude (Theorem 14.4) and the strand weights (Theorem 13.4), now on the vanishings themselves.*

Theorem 18.2 (The purity engine; `tensor_power_purity`, `duality_forces_purity`, `purity_engine`). *(i)* Weil I's square-root trick*: if every even amplitude power obeys a positive-pairing bound, $\|\alpha\|^{2k} \leq C \cdot B^k$ for all k, then $\|\alpha\|^2 \leq B$ — the constant dies under the k-th root. (ii)* The functional-equation endgame*: if additionally $\alpha\beta = B$ (the det-one pairing) and both one-sided caps hold, then $\|\alpha\|^2 = \|\beta\|^2 = B$ exactly — equality, i.e. purity, exactly as Weil I closes. Chained (`purity_engine`): det-one duality plus tensor-power bounds on both strands force both eigenvalue amplitudes to the half-unit $\sqrt{B}$.*

Theorem 18.3 (The Castelnuovo reduction; `cup_growth_gives_tensor_bound`, `purity_from_castelnuovo`). *If a vanishing class $v \neq 0$ is an eigenvector of the height transport T with eigenvalue α, and T's cup-norm growth on v is at the half-unit rate —*

$$\|T^k v\|^2 \;\leq\; C \cdot B^k \cdot \|v\|^2 \qquad (\text{the helix Castelnuovo inequality}),$$

with B the carrier's own area-law rate — then the engine's hypothesis holds for α; with the growth bound on both strands, every corresponding vanishing pair is pure (`purity_from_castelnuovo`). Everything except the growth hypothesis is proven; purity of every vanishing pair thus reduces to one *named operator inequality. Already proven beside it: the similitude sector has this growth as exact* equality *(`transport_cup_modulus`), and the $k \to \infty$ limit form is von Neumann reality (`von_neumann_reality`). Measured [tmp/castelnuovo_test.py]: on* 30 *vanishing classes and* 30 *random heights, $C_k \leq 1$ for every class and every $k \leq 4$ — the inequality holds with $C = 1$ as an* identity of the carrier metric*, and zeros are indistinguishable from random heights: the purity structure is built into the helix geometry (the $\sqrt{m}$ Frobenius weight cancels the $m^{-1/2}$ amplitude exactly), not conjured at the vanishings. The open mathematics is completion membership, not the growth bound.*

Theorem 18.4 (The conjugation axis is the membership boundary; `conj_axis_is_membership_boundary`, `carrier_strand_dominance`). *The readout class at abscissa σ is square-summable — lies in the Hilbert space the cup form completes to (`hilbert_completion_exists`) — iff $\sigma > \frac{1}{2}$: the half-unit is the edge of the Hilbert space itself. A hypothetical off-axis vanishing class would live* inside *the space, where the cup-symmetric flow has real spectrum and the Hermite–Biehler exclusion applies; the actual vanishing classes sit exactly* on *the boundary, where resonance is permitted. The 3D proof shape is therefore an exclusion — membership plus self-adjointness forbids off-axis vanishings — and it never needs to place the on-axis classes inside. The Hermite–Biehler mechanism itself has a geometric origin: per term the helix strand weight dominates the antihelix weight exactly for $\sigma > 1$, $r^{2-\sigma} < r^{\sigma} \iff \sigma > 1$ for every carrier radius $r > 1$ (`carrier_strand_dominance`) — the area law's growing radius* is *the strict HB inequality, with equality precisely at $\sigma = 1$ (the fixed point of the strand-swap $\sigma \mapsto 2 - \sigma$; the conjugation axis $\sigma = \frac{1}{2}$ enters through the completed weld, not this per-term bound).*

Theorem 18.5 (Clock-RH: the purity-defect law, unconditional; `ChiralityHB`). *Symmetrize each prime clock of the carrier at the half-unit: $E_{\alpha,\ell}(z) = e^{iz\ell/2} - \alpha\, e^{-iz\ell/2}$, with $\ell > 0$ the winding rate (*chirality*) and α the clock face. A zero forces the full winding onto the face (`symClock_zero_iff`),* *and its height obeys the exact law*

$$\operatorname{Im} z \;=\; -\frac{\log \|\alpha\|}{\ell} \qquad (\texttt{clock\_zero\_depth}):$$

the displacement off the conjugation axis equals the purity defect of the clock face — *purity and axis-location are the same statement, pointwise at every clock. Hence for the proven determinant-one (unitary) clock faces, every zero of every clock is exactly on the conjugation axis (`clock_zeros_real`), and the carrier — the FTA product of its prime clocks, which never collide (Theorem 12.4) — has all zeros on the conjugation axis: `carrier_zeros_real`,* RH for the multiplicative carrier as defined, unconditional. *The statement is quantified over* arbitrary finite families *of unitary clock faces and winding rates — every character and every Satake configuration at once: GRH-shaped generality at the carrier level. The unitary clock is moreover a* real *de Branges function: $E^* = -\overline{\alpha}\, E$ (`symClock_star`), so $\|E^*\| = \|E\|$ identically (`symClock_selfdual_modulus`) — it lives exactly on the Hermite–Biehler boundary, where real zeros are required. Honest scope: this closes the* multiplicative *side, the carrier and its medium; the additive interference — the summed fiber's closures, the L-vanishings — is the de Branges completion question (Theorem 18.4, Remark 22.2), open and named.*

Remark 18.6 (The two walls are one — and neither is a 3D object)**.** *The purity program's open input was attacked under a design-time circularity audit, and the audit reshaped the program. The class definition passes: a vanishing class is the fiber state at a closure event — defined by closure at a real height, no* $\operatorname{Re}\rho$ *anywhere. The trap sits one step later: `purity_from_castelnuovo` needs the class to be an eigenvector* inside *a normed space, and Theorem 18.4 makes Hilbert membership of readout classes* equivalent *to $\sigma > \frac{1}{2}$ — so "prove the growth bound on vanishing classes in H" would be RH-in-disguise for the on-axis classes, which are boundary objects and never in H. (The measured Castelnuovo of Theorem 18.3 — an identity of the carrier metric, zeros indistinguishable from random heights — had already hinted the content does not live in the growth bound.) The resolution: the membership dichotomy* is *the proof shape. A hypothetical off-axis vanishing datum defines a class* inside *the completion, where the cup-symmetric flow has real spectrum (`von_neumann_reality`) and Hermite–Biehler excludes it (`hb_no_zero_upper`); on-axis classes live on the boundary, where resonance is permitted. Nothing needs to be included — off-axis needs to be* excluded*, and membership does that, given one hypothesis: that the summed fiber's structure function is Hermite–Biehler. Completion membership and summed-fiber HB are therefore* one wall*: the fiber's winding coherence.*

The kernels around that wall are proven (`SummedFiberHB`): the star is additive (`estar_sum`); products of boundary functions stay on the boundary (`product_stays_boundary` — with Theorem 18.5 the multiplicative sector is closed); a common weld phase keeps a sum exactly on the boundary (`common_weld_sum_stays_boundary`); and phase alignment with margin forces the strict *HB inequality for the sum (`aligned_strict_sum_HB` — alignment defeats the triangle inequality, and the named failure mode, big strands cancelling while small strands conspire, is exactly what coherence must forbid). The whole remaining wall is then welded into a single implication (`coherence_implies_conj_axis`):* winding coherence $\Rightarrow$ the summed fiber is Hermite–Biehler $\Rightarrow$ every zero of its collapse wave is real. *One geometric hypothesis — where de Branges spent decades — and everything after it proven. Its measurable face, the per-term alignment margin across the bank, is pre-registered in Section 20.3.*

The frame reading is the sharpest form of the split: the upper half-plane, the strip, the σ-coordinate, and the coherence hypothesis are all 1D-chart objects *— in the 3D representation the walls are not even expressible. The 3D package carries no hypothesis at all (`helix3D_RH`, Theo-*

rem 18.7); the 1D chart — the chart with the extra dimension — carries exactly the two conditionals it was always going to carry: the bridge (`projection_complete_iff_RH`, which is *classical RH) and the coherence hypothesis of `coherence_implies_conj_axis`. Honesty check, per the house rules: HB-of-the-sum is GRH-strength but neither assumes GRH nor is circular — it is a strand-dominance inequality about the function, with zeros-on-line as consequence, not premise.*

Theorem 18.7 (The frame split; `ProjectionCompleteness`). *The "upper half-plane" is not a 3D notion: the half-plane, the strip, and the σ-coordinate belong to the 1D projection chart. A 3D vanishing datum is a* real height γ, *and its image in the 1D chart has abscissa exactly $\frac{1}{2}$ by the projection map (`helix_vanishing_projects_to_conj_axis`, `arisesFromHelix_on_conj_axis`; the named capstone is `helix3D_RH`) — not by a theorem about analytic continuation. And classical RH for the completed zeta is proved* equivalent *to the statement that every zero of the 1D continuation arises from a 3D vanishing:*

$$\big(\forall\rho,\ \Lambda(\rho)=0\Rightarrow\rho \text{ arises from the helix}\big) \iff \big(\forall\rho,\ \Lambda(\rho)=0\Rightarrow \operatorname{Re}\rho=\tfrac{1}{2}\big) \qquad (\texttt{projection\_complete\_iff\_RH}).$$

The positioning claim of this paper — RH $\iff$ the projection is complete *— is a theorem. It is a* reframing*: "arises from the helix" is* defined *as "is the projection of a real height at which Λ vanishes," so the equivalence is obtained by unfolding the projection map, not by any analytic input, and no analytic content is smuggled in or out. What it accomplishes is exactly the relocation of the open question — all of the difficulty sits, undiminished, in the antecedent (*is the projection complete?*), which is classical RH itself; the theorem asserts nothing about that antecedent.*

Remark 18.8 (The stance: a theorem in an expanded frame). *The reading is exactly parallel to Weil and Deligne over $\mathbb{F}_q$: there, "RH" was proved unconditionally for a different, geometrically defined frame, and the classical statement remained a separate question. Here: for the 3D frame* as defined*, RH is unconditional and kernel-checked — the carrier's clock-RH (Theorem 18.5), the vanishing data projecting only to the conjugation axis (Theorem 18.7), the duality pairing with dimension symmetry and det-one blocks (Theorem 18.1), with purity of the vanishing pairs reduced to the single Castelnuovo growth inequality, measured as an identity (Theorem 18.3). What remains open is a statement about the 1D chart alone:* projection completeness *— the name we use henceforth for the pivotal hypothesis of Remark 17.7, whose per-character pencil form is `EveryZeroHasSource` (every zero has a source; its negation, a* sourceless *zero) and whose completed-ζ form is `projection_complete_iff_RH`. It is never claimed; Section 20 weighs the evidence on it.*

19 Computational phenomenology

The preceding sections are theorems. This section is science: the model run as an instrument, with its measurements, its error budget, and its negative results. All quantities below are reproduced by the repository harnesses (`phasor_explorer/model.py`, `focal_closure.py`, `carrier_fiber/carrier_fiber.py`) and the interactive WebGL explorer (`phasor_explorer/explorer_v2.html`), which renders up to 10^5 phasors with shader-evaluated phases and exposes every mode discussed here.

19.1 The partial-absorption law

The fiber starts empty at the carrier origin (the $n=0$ site carries nothing; the bank lives at heights >0), and each phasor is absorbed *continuously*: with head height H, phasor n carries the window

$$a_n(H) \;=\; n^{-1/2}\,\mathrm{clip}\big(H-n+1,\,0,\,1\big),$$

so it starts growing as the head leaves site $n-1$ and completes exactly *at* its own integer site — one phasor partial at any instant, nothing discontinuous. At integer heads the absorbed bank equals the completed snapshot exactly, and between sites the signed sum moves along a straight segment whose velocity is the single absorbing phasor's term (α-linearity, verified to 10^{-12}). One-at-a-time absorption is load-bearing: it keeps the eigen-heights strictly monotone, which is the hypothesis chain of Theorem 16.1. A τ-window variant $a_n(H) = n^{-1/2}\mathrm{clip}((H-n)/\tau, 0, 1)$ — many phasors growing simultaneously — remains available as a smoothing device; it sharpens the focal residual (below) at the cost of the one-at-a-time structure.

19.2 Validation: all critical-line zeros below height 36, eight characters

For the eight real primitive characters of conductor $q \le 12$ (the η-regularized principal channel and the quadratic characters of conductor $3, 4, 5, 7, 8, 11, 12$) we computed *every* critical-line zero with $\gamma < 36$ — 111 ordinates in total ($5, 11, 12, 14, 16, 16, 18, 19$ per conductor) — by Newton refinement on $L(s,\chi) = q^{-s}\sum_a \chi(a)\zeta(s, a/q)$ (Hurwitz-zeta evaluation, 22 significant digits, stored to 10 decimals). The bound $\gamma < 36$ is an integer-exactness cap, not a cost cap: the eigenheight $N = \mathrm{round}(e^{\gamma})$ must stay below 2^{53} for exact modular arithmetic.

At every one of the 111 pairs (γ_k, $N = e^{\gamma_k}$) the focal event of Theorem 15.2 was evaluated. With the absorption law (equivalently the completed snapshot, since N is an integer) the residual $|B|$ lies below the display threshold 0.05 at 108/111 zeros; the three exceptions are precisely the smallest banks — conductor 7 (γ_1, $N = 88$, $|B| = 0.107 \approx 1/\sqrt{88}$), conductor 8 ($N = 134$, 0.087), conductor 12 ($N = 45$, 0.153) — honest truncation at that scale, and the τ-window closes all three (0.008–0.016 at $\tau = 400$). At the deep end the residuals sit at the truncation scale $\sim N^{-1/2}$: for conductor 11, $\gamma_{18} = 35.4850851876$ gives $N = 2{,}576{,}182{,}862{,}426{,}122 \approx 2.6 \times 10^{15}$ and $|B| = 7.3 \times 10^{-11}$; for ζ, γ_4 gives $N \approx 1.6 \times 10^{13}$ and $|B| = 1.2 \times 10^{-7} \approx N^{-1/2}/2$.

19.3 The error budget

The residual at a focal event decomposes into three measurable floors. *Zero precision*: with ordinates stored to 6 decimals the deep residuals are dominated by the rounding of γ itself (conductor 11, γ_{18}: 6.2×10^{-7} at 6 decimals versus 7.3×10^{-11} at 10), so headline depths quote the data precision, not the model. *Truncation*: with the zeros at 10 decimals the residual tracks $\sim N^{-1/2}$ across ten orders of magnitude of N. *Evaluator*: beneath both lies the analytic evaluator's own floor ($\approx 10^{-9}$–10^{-10}, next subsection). Misattributing the first floor to the second is the kind of error this budget exists to prevent.

19.4 The $O(1)$ evaluator

Brute evaluation is $O(N)$ and the eigenheights reach 10^{15}. Every phase mode except the continuous carrier-rate comparison has residue-class structure, so the bank sum splits as an exact 6000-term head plus per-class Euler–Maclaurin tails (integral, boundary, and B_2, B_4, B_6 corrections, all closed-form for n^{-s}), with the absorption window contributing one explicit partial term. The cost is independent of N; the absolute error is $\sim 10^{-9}$. The evaluator is validated against the brute sum on 240 configurations (eight characters $\times$ three phase modes $\times$ five magnitude laws $\times$ two sizes, agreement $< 10^{-8}$) plus a 6.6×10^{6}-term brute anchor, and the JavaScript port agrees with the Python reference to nine decimals at fractional heads. This is what makes Section 19.2 a table instead of a supercomputer allocation.

19.5 Degree two: the mechanism does not require periodicity

The cheapest dismissal of Section 19.2 would be that character periodicity — complete residue blocks cancelling — secretly does all the work. It does not. The locator of `focal_closure.py` runs the same construction on degree-two L-functions, with the lanes given by *coefficient signs* instead of residue classes: for the Ramanujan form, $\lambda(n) = \tau(n)/n^{11/2}$ built exactly from $\eta(q)^{24}$ (with Hecke multiplicativity and Deligne-bound assertions checked at build time), and for the level-11 elliptic newform, $\lambda(n) = a_n/\sqrt{n}$ from $\eta(q)^2\eta(q^{11})^2$. The growth locator finds the first zero of $L(\Delta, s)$ unaided to 10^{-6} of the reference ordinate $9.2223793999\ldots$, with genuine closures reaching depth 10^{-6}–10^{-12} as the bank deepens; the level-11 zeros likewise. Neither form is periodic, neither has unimodular coefficients. Within the arrows the $\pi/3$ scale is provably inert (a constant amplitude and a global phase, cancelling between lanes — the gauge test of `focal_closure.py`); its load-bearing roles are exactly the two proved ones: the carrier placement forcing the area law (Theorem 3.9) and the exact cell normalization $B = (\pi/3)L$ (Theorem 15.2).

Satake silence. The ambient spectroscopy of Section 19.7, run on the degree-two forms (`tmp/gl2_explore.py`), reads the local Satake parameters: the clock line at $\ln p^2$ carries amplitude $\|\alpha^2 + \beta^2\|/2$ for the unitary pair $\alpha + \beta = \lambda(p)$, $\alpha\beta = 1$. Newton's identity gives $\alpha^2 + \beta^2 = \lambda^2 - 2$ exactly (`satake_newton_two`), so the p^2-line is silent *iff* $\lambda(p)^2 = 2$ — Satake angle $\pm\pi/4, \pm 3\pi/4$ (`satake_silence_iff`). Both signs of the prediction are measured: for Δ, $\lambda(2)^2 - 2 = -1.72$ and the $\ln 4$ line is its second strongest (0.71 relative to $\ln 2$, an $8.5\times$ enhancement over the degree-one pattern); for the level-11 curve at $p = 2$, $a_2 = -2$ gives $\lambda(2) = -\sqrt{2}$, hence $\alpha^2 + \beta^2 = 0$ exactly (`e11_ln4_silent`) — measured at 0.0012 relative power: the line the spectroscopy found dead is dead by algebra, a measured null promoted to a theorem. The supersingular silences are as sharp ($\ln 19 \leq 9.4 \times 10^{-7}$, $\ln 29 \leq 3.5 \times 10^{-8}$). A long-span sweep quantifies the law: across $\sim$40 detected lines of Δ, the level-11 curve, and a CM control form, measured versus predicted Satake weights agree to $\pm 4\%$ (medians 1.007, 1.051, 0.994), including the alternating 2-power tower of the CM form verified tooth-by-tooth at its exact inert angle; the CM lattice angles themselves ($\pi/2$, $2\pi/3$) carry no clock — a clean geometric-clock null (record: `tmp/gl2_longspan_results.txt`). The $\pm 4\%$ envelope proved to be method, not medium. Regenerating the tables with the calibrated, unclipped estimator [no clip of $\log|F|$; windowed projection at parabolically refined peaks; a per-run calibration on the exact truncated explicit series that reads 1.00000 by construction; bank $N = 1.5 \times 10^6$, $t \in [100, 2200]$; record `tmp/satake_clean_results.txt`, driver `tmp/satake_clean.py`] the degree-two weight law holds to $\sim$1%: over the calibration-gated $k = 1$ lines, Δ has median $0.9997 \pm 1.1\%$ and the level-11 curve $0.9636 \pm 1.0\%$, against the old $1.007 \pm 4.5\%$ and $1.051 \pm 3.1\%$. Two systematics are isolated and reported rather than absorbed: (i) the residual is a *finite-bank* line attenuation, not the estimator — the level-11 meas/pred lifts $0.66 \to 0.83 \to 0.95$ and the detection floor sinks $1.4 \times 10^{-2} \to 9.9 \times 10^{-3} \to 3.2 \times 10^{-3}$ as N runs $2 \times 10^5 \to 8 \times 10^5 \to 1.5 \times 10^6$ (the real-line signature); (ii) the $k = 3$ lines are intermodulation-contaminated (the weak $3 \ln p$ bins collect $\ln p \pm \ln q$ mixing power, inflating meas/pred 3–500$\times$) and are excluded from the law and reported separately — the calibration column flags exactly these, along with a handful of very weak $k = 2$ lines whose search window snaps to a neighbour. The $\ln 4$ Satake hole and the supersingular $\ln 19, \ln 29$ silences all sit at the floor (1.245, 0.925, 0.467 times floor) as predicted.

19.6 Negative results

Three natural conjectures about the carrier were tested and are null, and one phenomenological law is positive. (i) A $\pi/3 \cdot \mu_6$ carrier-pattern search for *close pairs* of zeros shows no enrichment:

lift ≈ 1.0, $p \approx 0.9$. (ii) A pair-precursor signal (structure in the fiber preceding a close pair) is absent. (iii) A 6- or 12-fold "harmonic clock" in the zero sequence is absent; only the $k = 1$ harmonic is real. (iv) Positive: after each cancellation the fiber's imbalance *reopens* at a fixed linear rate ($p \approx 1.05$, uncorrelated with the gap to the next zero), with close (Lehmer-type) pairs entering through a quadratic crossover rather than any repulsion mechanism; the reverb kernels are formalized in `ReverbResidue.lean`. We report (i)–(iii) with the same confidence as the positive results: the carrier structure is a home for the cancellation event, not a hidden periodicity in the zeros.

19.7 The ambient field: Euler clocks and conductor holes

Between the zeros the fiber is not featureless. The power spectrum of $\log|L(\frac{1}{2} + it)|$ (`tmp/ambient_spectrum.py`) shows sharp lines at *every* $\ln p^k$ — for ζ the $\ln 2$ line stands 2.5×10^6 over background, with weights falling as the Euler expansion predicts — and exact nulls at composite non-prime-powers ($\ln 18$, $\ln 21$: consistent with noise). The conductor is visible as a *hole*: the complex quintic character's ambient is missing exactly its $\ln 5$ line and the mod-29 character's exactly $\ln 29$ (noise-level, versus 6.5×10^5 and 6.6×10^4 for ζ), every other prime matching — the ramified prime is the neutral bucket ($\chi(p) = 0$: no arrow at p, hence no p-clock in that fiber), made spectroscopically visible. The division of labor is measured on both sides: the *configuration* observables unfold — the mid-gap reopening amplitude collapses onto a single smooth curve $b(g) \approx 1.35\, g^{2.3}$ in unfolded units across four heights, three conductors, real and complex characters, and both degree-two forms (collapse score 0.137 unfolded versus 1.850 raw) — while the *medium* carries the raw $\ln p$ clocks. Only the Euler log-clocks exist: every π-multiple tested ($\pi/3$, $\pi/2$, $2\pi/3$, π — including each CM field's own lattice angle) is null, the one apparent hit at $4\pi/3$ resolving to $\ln 67$ at $\Delta\omega = 0.0018$; bounds $\leq 3.4 \times 10^{-6}$ (32.a $\ln 4$-line, relative to its $\ln 5$ power) and $\leq 8.1 \times 10^{-4}$ ($\pi/2$) [`tmp/pi_clock_results.txt`]. Two structural theorems stand behind the spectrogram: the lines *are* the carrier's sites recovered by Bohr inversion (Theorem 12.3), and no two lines can ever collide, because $\log p/\log q$ is irrational for distinct primes — unique factorization as spectral non-collision (Theorem 12.4). The conductor axis substitutes cheap heights for expensive ones: χ mod 29 at $t \approx 1875$ (mean spacing 0.693) reproduces ζ-at-60,000 behavior. The ambient texture is the readout shadow of the FTA winding (Theorem 4.2: multiplication becoming angular addition, prime angles reaching $t \ln p$ through the bridge), and it is where the arithmetic lives that the unfolded zero statistics forget.

A full census closes the chapter [`tmp/ambient_memory.py`; ζ and the complex quintic character, $t \in [1000, 3000]$]: $81+79$ in-band lines, every one at some $\ln p^k$; after notching all of them, every surviving residual peak is an *intermodulation product* $\ln p \pm \ln q$ (lock-and-grow verified; at 10^{-5}–10^{-6} of the $\ln 2$ line power, versus 10^5–$10^6\times$ local median for the primaries) — nothing off the additive group generated by $\{\ln p\}$, with π-multiple bounds $\leq 6.5 \times 10^{-5}$ of the $\ln 2$ power. *The ambient spectrum is supported on* $\log \mathbb{Q}_+^\times$*: unique factorization is the medium's entire memory.* The one loose end — an apparent autocorrelation excess for the quintic character — resolved as a surrogate-class artifact, and the resolution corrected our own first reading [`tmp/acf_surrogates.py`]: IAAFT surrogates are circular for an ACF statistic (they fix the spectrum, whose transform *is* the ACF; the real ACF equals its own surrogate band to 10^{-4} — zero memory excess); the marginal explains nearly nothing, refuting our dip-clipping suspicion (0.009 against the real 0.34); the colored residual spectrum — Euler-line skirts, intermodulation lines, clip ripple — explains all of it; and the lag profile is sub-spacing (0.3–0.4 mean spacings, monotone decay): the deterministic reverb arch of Section 19.9, with nothing at the multi-spacing lags where genuine cross-zero memory would live. A configuration-echo test at 1/2/3 mean spacings sits within the null.

19.8 Functoriality, strand phases, and the counting statistic

Functoriality as measurement. The Gelbart–Jacquet lift $\mathrm{Sym}^2\Delta$ — a genuine GL(3) L-function — is built from the exact $\tau(n)$ via the $(\lambda^2 - 1)$ ladder [`tmp/sym2_spectroscopy.py`; span 2200, bank $N = 800{,}000$]: construction certified against the Euler product at 5.6×10^{-11} (at $s = 2$), with the GL(3) unitarity bound $|c_{p^k}| \le \binom{k+2}{2}$ holding exactly. Its medium obeys the *lift's* weight law. The historical figure was measured/predicted 0.82 ± 0.13 over 18 clean lines, with the 0.82 traced to the degree-3 continuum floor by a matched ζ control (10^3–10^4 times higher floor at equal span). Clean regeneration confirms this diagnosis exactly and sharpens it [`tmp/satake_clean_results.txt`, bank $N = 1.5 \times 10^6$]: because the degree-three approximate functional equation scale grows as $(t/2\pi)^{3/2}$, a fixed bank supports a lower maximum height for GL(3) than for GL(2). A height ladder makes this quantitative — the $k = 1$ meas/pred median runs 1.0006 ($t \le 500$) $\to$ 0.9367 $\to$ 0.8244 $\to$ 0.8297 ($t \le 2200$), degrading monotonically as the height outruns the bank while the calibration column stays pinned at 1.00000. *At the height the bank fully supports* ($t \le 500$) the degree-three weight law reads $1.0006 \pm 0.2\%$: the same functoriality figure as degree two, once the finite-bank attenuation is removed. The 0.82 at full span is the undersized-bank artifact, not a degree-three defect. The decisive test is two-sided, and every decisive prime follows the lift: at $p = 43, 73$ ($\lambda_p \approx 0$ — the base's own law predicts silence, the lift predicts bright) the lines are measured *loud*, 50×-plus over the base prediction; at $p = 11, 19$ ($\lambda_p^2 \approx 1$: the Eisenstein angle $\theta_p = \pi/3$) — base loud, lift silent — they are measured *dead*. Shape correlation across 25 primes: +0.36 for the lift, −0.23 for the base (the wrong ordering). Composite nulls hold; line locations lock below 0.001 rad. This is the Langlands reading's queued experiment, run (Section 21). The vanishing side followed: locator, cells, pinning, and census, all exact at degree three (Section 19.11).

The medium is strand-aware. The *complex* spectrum of $\log F$ carries the oriented datum in its line phases [`tmp/strand_phase.py`]: with the convention calibrated on ζ (all phases 0 ± 0.013 rad) and then frozen, the order-4 character mod 5 reads $\arg \chi(p^k)$ exactly ($\ln 2$ at $+0.491\,\pi$ for $\chi(2) = i$; all quadrants correct, span-locked), and Δ, E11 read sign λ_p (all eight lines in the correct half-plane); ramified holes wander — correctly no line. Power spectra see the Satake modulus; the phases see the class itself.

The counting statistic. The fiber-native $S(t) = N_{\text{fiber}} - \theta/\pi - 1$ (exact theta, no L-calls) [`tmp/st_cells.py`]: the number variance $\Sigma^2(L)$ tracks GUE (exact form) in the rigid regime and saturates at Berry's outer scale $\log(t/2\pi)/\pi$, the saturation moving outward with height (departure at 2.4 spacings for $t \le 2000$, 3.4 at $t \approx 51{,}000$) — so the counting and rate observables saturate on the *same* clock $\log(t/2\pi)$ as Section 19.9's dictionary: one clock for both. Selberg's variance constant is stable at $C \approx 1.43$ (sd 0.017). Honesty note: over this height range $\ln \ln(t/2\pi)$ spans only about one unit, so the log-log growth itself is unresolved here; and S is a sawtooth, not yet Gaussian (KS 0.0146 against the normal on a dense grid).

19.9 The reverb law and the CUE bridge: no repulsion required

The positive reverb finding of Section 19.6(iv) has a formal kernel, and that kernel turns out to be *shared, identically, with random-matrix theory* — which recalibrates what GUE-style agreement can and cannot evidence.

The reverb kernel. At a vanishing $f(\rho) = 0$ of a differentiable fiber, the local slope reopens at the derivative: $f(y)/(y - \rho) \to f'(\rho)$ (`reopening_slope`); a nonzero reciprocal-fiber residue exists iff the vanishing is simple (`residue_exists_iff_simple`); and near a cluster S of other vanishings the reopening rate carries the exact distance product,

$$\|f'(\rho)\| \;=\; \Big(\prod_{\sigma\in S}\|\rho-\sigma\|\Big)\cdot\|h'(\rho)\|,$$

with h the regular cofactor, proven analytic across the partner (`cluster_product_law_norm`, `pair_suppression`, `pair_regular_across_partner`). This is the law measured in Section 19.6(iv), and the measured record is quantitative [`tmp/reverb_clusters.py`, `tmp/reverb_profiles.py`, `tmp/gourdon_pair_test.py`]: $\hat{R} \propto S^{\beta}$ with $\beta = 1.01$ over the 1517 zeros below $t = 2000$ (zero count matching Riemann–von Mangoldt, 1517 vs 1516.6); suppression compounds at triples (isolated 0.96, pair member 0.57, middle of the tightest triple 0.32); the inter-zero profile is a symmetric arch with peak fraction $\varphi = 0.44$–0.49 (Lehmer pair 0.476), triple peak ratios matching the parameter-free product model to 2–6%; and the law persists from $g = 0.5$ down to $g = 8.2\times10^{-5}$ at the Gourdon pair at $t \approx 1.085\times10^{10}$ ($\hat{R}/g = 2.93$, arch $\varphi = 0.500$ exactly, while a fixed-slope tent misses by $8300\times$). Suppression by bookkeeping, with no repulsion dynamic anywhere in the mechanism.

The CUE bridge. The random-matrix "derivative statistic" is the *same object.* For the characteristic polynomial of any complex matrix, the derivative at an eigenvalue is exactly the product of distances to the remaining eigenvalues (multiset form, multiplicity-safe: `derivative_prod_roots_eval`, `cue_derivative_at_eigenvalue`), in norm $\|P'(\mu)\| = \prod\|\mu-r\|$ (`cue_rate_eq_distance_product`); and the characteristic polynomial, viewed as a fiber, satisfies the *identical* reopening law — the formal statement `cue_reopening_slope` is proved by applying the fiber kernel `reopening_slope` to the matrix side. One mechanism, two universes.

Consequence, stated carefully. Agreement between L-zero derivative statistics and CUE/GUE predictions in reopening rates, cluster arches, or close-pair suppression is an *identity of shared bookkeeping* — both sides compute the same distance product — and is therefore not, by itself, evidence of a repulsion mechanism acting on the zeros; statistical differences can enter only through the configurations themselves [13]. Measured against CUE with the Keating–Snaith dictionary $N = \log(t/2\pi) \approx 6$ — large N is the wrong dictionary here; $N = 60$ introduces a spurious $3\times$ spread — the raw comparison gives KS = 0.114 and the cluster-corrected one KS = 0.066 [`tmp/reverb_clusters.py`]: the derivative statistics are carried by the gap configuration through the product law. Nothing in this subsection assumes or asserts anything about where the zeros lie; the theorems are unconditional and machine-checked (`ReverbResidue.lean`, `GUEBridge.lean`).

19.10 The reverb tax inverted: the vanishings carry the clocks

The $\pm4\%$ envelope of Section 19.5 was interrogated in three steps, and what lay underneath inverted our hypothesis.

(i) Structure, not noise [`tmp/satake_variance.py`]: split-span residuals reproduce at cross-span correlation $+0.969$ (noise predicts ≈ 0), so the variability is systematic. *(ii) Method, then calibration* [`tmp/reverb_tax.py`]: with a windowed-projection estimator calibrated at 1.00000 on the exact series, the residual becomes a reproducible, all-negative deficit of $\approx$1%; the no-clip control then removes it entirely (mean $+0.0003$ unclipped) — the deficit was the amplitude clip truncating the dip cores. Unclipped, *the ζ Satake weight law holds at* $\sim$0.1%. (The historical $\pm4\%$ of the long-span GL(2)/GL(3) tables was span, floor, clip, and undersized-bank method — not medium. Clean regeneration [`tmp/satake_clean.py`, bank $N = 1.5\times10^{6}$; Sections 19.5, 19.8] confirms this: degree two holds to $\sim$1% (Δ 0.9997 $\pm$ 1.1%, level-11 0.9636 $\pm$ 1.0% on calibration-gated $k = 1$ lines) and degree three to $\sim$0.2% at the height its bank supports, the residuals resolving into a finite-bank line attenuation and $k = 3$ intermodulation, both reported.) *(iii) The inversion.* We

asked whether the dip neighborhoods — the vanishings — *resist* the prime clocks: a reverb tax. The opposite is true: excising the dip neighborhoods (11% of the span around the 1820 located vanishings) removes ≈10% of every clock's coherent line power, growing with $\ln p$ (-2.6% at $\ln 2$ to -13.6% at $\ln 37$). The dips *carry* the clocks.

The inference is kernel-precise [`ClockDipDuality.lean`]: a dip of common profile at position γ contributes to the clock of frequency ω with phase $e^{i\omega\gamma}$ — position becomes phase, identically (`dip_projection_phase`, translation covariance); if the positions are phase-locked to the clock, J dips produce line amplitude scaling with J (`locked_dips_add`); if they march at any off-resonant spacing, the phase sum is bounded independently of J (`unlocked_dips_cancel` — proven for the equal-spacing model case; the measurement supplies the generic-position contrapositive). Measured: J-scaling coherence at *every* clock $\ln p$ simultaneously. By the dichotomy, the vanishing set is phase-locked to every prime clock — the Riemann–Weil explicit-formula duality, the zeros knowing the primes, observed as spectroscopy. And a method law, recorded the honest way: *never clip when measuring line amplitudes* — the dip cores are signal.

19.11 Exact phase quantization: the fiber as its own oscillator

The phase side of the cancellation cell quantizes exactly — a measured result whose two supporting kernels are now proven (Theorem 13.1); we mark the tiers explicitly. Raw measurement [`tmp/phase_cells.py`]: the fiber's phase flips by 0.98–$0.99\,\pi$ at every vanishing, winds $\approx -\pi$ between vanishings, and reaches its first vanishing at $|\Phi| \approx 0.55\,\pi$. The fluctuations are the readout's, not the model's: correcting each cell by the *running carrier gauge*

$$\theta'(t) \;=\; \tfrac{1}{2}\ln\frac{qt}{2\pi} \;=\; \ln\!\big(\text{carrier radius at the resonant site}\big)$$

— the pure $\sqrt{n}$ radius, exactly in the $\pi/3$ unit gauge, so the area law (Theorem 3.9) acts as the phase gauge — removes 99.2–99.5% of the cell fluctuation: every cell is exactly π with standard deviation 0.001–$0.002\,\pi$, for ζ and all four tested characters [`tmp/s_running.py`]. The classical contact is Hardy-Z reality; the model-native content is that the theta clock *is* the log of the bank radius, and the fluctuation term $S(T)$ never appears in the carrier's own chart. Passing to the oscillator frame — the analytic signal of the de-chirped fiber, extended *evenly* across the hinge, where the anti-helix conjugacy makes the hinge a turning point (real, even, zero velocity) — the quantization becomes exact: first crossing at 0.499–$0.500\,\pi$ (a quarter cycle), every subsequent cell 0.999–$1.000\,\pi$ (half cycles), standard deviation down to 0.001 [`tmp/oscillator.py`]. Zero counting is then deterministic,

$$N(T) \;=\; \Big\lfloor \frac{\varphi(T)}{\pi} + \frac{1}{2} \Big\rfloor,$$

with no theta asymptotics, no $S(T)$, and no additive constant — the classical "+1" is revealed as the hinge quarter-cycle. *Honest decomposition*: given (i) Hardy reality (measured at $0.002\,\pi$), (ii) evenness of the hinge extension, and (iii) simplicity of the vanishings, the quanta are structural; (ii) and the evenness→turning-point step, pending at first writing, are now *proven* — `collapseWave_even` from one functional-equation line, and `hinge_turning_point'` as an honest `HasDerivAt` (Theorem 13.1). Dead ends are recorded with the same care: fixed gauges ($\pi/3$, $\pi/6$, ...) remove only 27–45% of the fluctuation at low heights — the cancelling gauge must *run* with height [`tmp/s_removal.py`, `tmp/phase_norm.py`]; the cell-integral fiber loses vanishing-fidelity (its zeros leave the L-ordinates); and the growth-path odometer undershoots the hinge (the gauge is undefined below $qt/2\pi = 1$).

Three extensions complete the phase story. *The root number is read from the line phase* [`tmp/root_hinge.py`]: with the exact Γ-phase gauge, the completed phase is constant between vanishings, jumps by exactly π at each vanishing, and is pinned to $\arg(\varepsilon)/2 \pmod{\pi}$ (Theorem 13.2) — measured $0.0881\,\pi$ for the order-4 character mod 5, $0.1868\,\pi$ for the order-6 character mod 7, conjugates mirrored ($0.9119\,\pi$, $0.8132\,\pi$), the real character mod 3 at 0 — each equal to the Gauss-sum $\arg(\varepsilon)/2$ to four decimals with zero circular spread. *The quanta survive degree two exactly* [`tmp/gl2_cells.py`]: Δ's cells are $1.0000\,\pi$ (sd $< 10^{-4}\pi$) and the level-11 newform's $1.0000\,\pi$ (sd $0.0012\,\pi$), both pinned to the $\varepsilon = +1$ ray at $\leq 10^{-4}\pi$ — the doubled winding lives entirely in the doubled Γ-gauge; the phase-cell law is degree-blind, as the pinning argument predicts. *And degree three exactly* [`tmp/gl3_vanishing.py`]: running the full vanishing-side machinery on $\mathrm{Sym}^2\Delta$ with the completed degree-three kernel ($\Gamma_{\mathbb{R}}(s{+}1)\,\Gamma_{\mathbb{C}}(s{+}11)$) gives 22 located ordinates independently confirmed at 25-digit precision, 21 consecutive cells at $1.0000\,\pi$ with sd $0.0000\,\pi$, census 15/15 simple, the $\varepsilon = +1$ turning point at the hinge, and a pinning spread of 4.0×10^{-14} — the tightest in the program. A bonus discovery: *the pinning discriminates the archimedean factor.* The wrong-parity Γ pins tightly (spread 7.6×10^{-5}) but onto a ray $\pi/4$ off ($0.2497\,\pi$ measured) — the Stirling constant of the parity swap, $\arg\big[\Gamma(\tfrac{1}{4} + it/2)/\Gamma(\tfrac{3}{4} + it/2)\big] \to -\pi/4$, exactly halfway between the two admissible weld rays — so a parity error can never masquerade as either root number. The fiber reads its own ∞-place local datum: the finite places by spectroscopy, the infinite place by the weld ray.

19.12 The weld read out: root numbers, the BSD rank ladder, and the dimension census

The hinge kernels of Section 13 have a quantitative experimental record on elliptic curves, run entirely on the model's own machinery: coefficients by point counting, completed lines by the two-strand incomplete-Γ kernel of Theorem 13.4, *no L-function library in the loop* [`tmp/bsd_weld.py`, `tmp/bsd_rank_ladder.py`, `tmp/rank4_weld.py`, `tmp/rank5_weld.py`].

The rank ladder 0–6. For 11.a, 37.a, 389.a, 5077.a, the rank-4 Elkies curve 234446.a (composite conductor $2 \cdot 117223$), the rank-5 Brumer–McGuinness curve $y^2 + y = x^3 - 79x + 342$ (prime conductor 19,047,851; bank to $n = 26{,}396$ by point counting), and the rank-6 Elkies–Watkins curve $[1, 1, 0, -2582, 48720]$ (conductor $N = 5{,}187{,}563{,}742$, four multiplicative bad primes; bank to $n = 435{,}598$ by point counting): the hinge jet tower shows parity-dead layers at 10^{-13}–10^{-16} (Theorem 13.3 in action) and rank-dead layers at 10^{-4}–10^{-9}, and *the first live jet is the BSD leading datum* — measured against $(\sqrt{N}/2\pi)\, L^{(r)}(1)/r!$ built from the G_r reference ladder (Theorem 13.5) out of our own coefficients:

$$\frac{\text{measured}}{\text{predicted}} = 1.00000,\ 1.00000,\ 0.99998,\ 0.99974,\ 0.99999,\ 1.00000,\ 1.00000 \qquad (r = 0, \ldots, 6).$$

Hinge value, reverb rate, curvature, jerk, quartic, quintic, sextic — rank is dimensions added at the hinge. At rank 6 the leading jet is $|c_6| = 3{,}677{,}144.76$ against the predicted 3,677,144.73, with dead layers at 3×10^{-17}–9×10^{-11} of leading, stable across four independent fit settings; the bad-prime signs are selected by a *triple* forced zero ($L(1)$, $L''(1)/2$, and $L^{(4)}(1)/4!$ simultaneously, at ratio 4.6×10^{-9}, three orders below the runner-up) — at $\sqrt{N} \approx 72{,}000$ the pinning read-off is fully masked, so the forced-zero test, not pinning, is the discriminator there, stated honestly. The same reference series reproduce Gross–Zagier's $L'(1) = 0.3059997738$ for 37.a and the BGZ rank-2/3 values to ten digits, produce the rank-forced zeros at ranks 4/5 at 10^{-7}–10^{-8}, and supply the record references $L''''(1)/4! = 8.943848$ and $L^{(5)}(1)/5! = 30.285687$. The large conductors run on a hybrid exact kernel (Gauss–Laguerre form of the incomplete-Γ strand) validated at 7.5×10^{-13}

per term against direct evaluation, with a full rank-4 regression agreeing at 1.000000. Two honest boundaries: the identification of $d(0)$ with the Mordell–Weil rank is BSD itself — measured tier only, never claimed as theorem; and the pinning read-off of the bad Euler factor ($a_{37} = -1$ at ray $0.5000\,\pi$, spread 4×10^{-7}; $a_{389} = +1$) works while $N^{-1/2}$ exceeds the single-strand fiber floor $\sim 10^{-2}$ — decisive at conductors 37 and 389, masked at 5077, where Atkin–Lehner theory supplies it instead.

The dimension census. Sam's dimension principle — every vanishing carries a dimension, $d(\gamma)$ = order of vanishing = dead depth of the local jet tower — measured across the ladder [`tmp/dimension_census.py`]: $d(\gamma) = 1$ at *all 31* non-central vanishings (the two flagged 5077.a cases adjudicated by the exact kernel at floors $10^{-15}/10^{-18}$, jet-to-floor $> 10^{11}$), while $d(0)$ = rank exactly (0/1/2/3). The ζ census is the same statement: 1517/1517 simple (Section 19.9). Only the weld point carries extra dimensions — the arithmetic deposits its points where the phase is pinned (Theorem 13.2).

Strand topology is a degree statement. This corrects our own first reading (a "large conductor" effect), and the correction is part of the record [`tmp/strand_topology.py`]: degree-one fibers are *hinge-outward* — the series converges conditionally on the line, so the single strand growing up from the midpoint is valid everywhere (Legendre character mod 10009: hinge error *decays* monotonically $5.7 \times 10^{-4} \to 5.2 \times 10^{-7}$ as the bank grows 50k$\to$ 800k) — while degree-two fibers are *endings-inward*: the series does not converge at the center, and the hinge exists only as the two incomplete-Γ strands starting at the cusps and meeting at the Fricke midpoint (5077.a: error *saturates* at ~ 0.3 at every bank size; the exact kernel self-checks at 5×10^{-12}). The conductor hypothesis is refuted by its own control — the character's resonant scale is five times larger, yet convergent. At the hinge both strands weigh equally (Theorem 13.4), which is why the antihelix cannot be smoothed away there: the single-strand bank floors at $\sim 10^{-2}$ by $N \approx 5000$.

19.13 Retractions and method

Two published-in-session readings were retracted, and the reasons are part of the record. *The growth-clock window* (a fixed-rate tent with horizon $\Delta y \approx 1$, one e-fold) was an artifact of two errors compounding: the η prefactor $|1 - 2^{1-s}|$ modulates amplitude data by $\pm 2.4\times$ with raw period 9.06 and must always be de-modulated, and mixed strata faked a saturation kink. On clean observables the skew statistic has *no* horizon (error decreases monotonically out to 5–8 spacings, `tmp/horizon_sweep.py`) and the amplitude curves collapse in unfolded units by $13.5\times$ over raw (score 0.137 vs 1.850, seven samples, `tmp/amp_cross.py`) — one universal curve $b(g) \approx 1.35\,g^{2.3}$, no saturation, degree- and conductor-independent. *The two "ghost lines"* (32.a near $\ln 4$, 27.a near $\ln 8$) were Hann-window skirt artifacts of strong neighbors; the resulting method law is now fixed: line claims require *local-maximum enumeration with parabolic interpolation*, never window-max ratios; a real line locks its location and grows with span while artifacts wander; blend tests use centroid pull and skirt symmetry; every anomaly gets a same-span control before it gets a name; and line-amplitude measurements are never clipped — the dip cores are signal (Section 19.10).

19.14 The finite instrument

The finite Python harness `carrier_fiber/carrier_fiber.py` separates the carrier geometry from the analytic calibration. Its native finite path uses the pitch-one $\pi/3$ carrier directly: a height $z > 0$ absorbs

$$N = \lfloor z \rfloor$$

complete integer slots and the fractional part $\alpha = z - N$ of the next slot. With a real periodic coefficient $c(n)$, the signed finite fiber term is

$$v_n = c(n)\,((\pi/3)n)^{-1/2}\exp\bigl(i(n-1)\pi/3\bigr).$$

The whole-fiber focal value at height z is therefore the piecewise-linear path

$$B(z) = \sum_{n=1}^{N} v_n + \alpha v_{N+1}.$$

The readout ordinate printed by the harness is $\log z$; the height itself is the carrier coordinate.

The native scan `find_continuous_focal_crossings` finds candidate vanishings without a one-dimensional nonlinear root solver. On each interval $[N, N+1]$ it writes $S_N = \sum_{n=1}^{N} v_n$ and minimizes the distance from the origin to the segment $S_N + \alpha v_{N+1}$ by the explicit projection

$$\alpha_N = \mathrm{clip}_{[0,1]}\left(-\frac{\mathrm{Re}(S_N\overline{v_{N+1}})}{|v_{N+1}|^2}\right).$$

The residual is

$$r_N = \bigl|S_N + \alpha_N v_{N+1}\bigr|.$$

Heights with r_N below the requested tolerance are reported as exact finite focal crossings; if no exact crossing is present in the scan range, the harness reports the smallest residuals. In compact form:

```
terms = c(n) * ((pi/3) * n)**(-0.5) * exp(1j * slot * pi/3)
prefix[N] = sum_{n <= N} terms[n]
alpha[N] = clip(-real(prefix[N] * conj(terms[N+1])) / abs(terms[N+1])**2, 0, 1)
height[N] = N + alpha[N]
residual[N] = abs(prefix[N] + alpha[N] * terms[N+1])
```

The same scan also evaluates the harmonic pencil marker. The unsigned admissibility channel is

$$A(z) = \sum_{n=1}^{N} |c(n)|\,((\pi/3)n)^{-1/2} + \alpha|c(N+1)|\,((\pi/3)(N+1))^{-1/2}.$$

For each pencil

$$H(z) = \begin{pmatrix} A(z) & B(z) \\ \mu A(z) & \lambda B(z) \end{pmatrix},$$

the code checks

$$\det H = (\lambda - \mu)A(z)B(z), \qquad \det(H^*H) = |\det H|^2.$$

Thus, whenever $A(z) \neq 0$ and $\lambda \neq \mu$, the Gram rank-drop marker is exactly the same event as the finite focal cancellation $B(z) = 0$.

The default validation command uses the cumulative c1 carrier clock, which is the Lean-aligned readout used for the L-channel:

$$T = \gamma/(\pi/3), \qquad B_N(T) = \frac{\pi}{3}\sum_{n\le N} c(n)n^{-1/2}\exp\bigl(-i(\pi/3)T\log n\bigr).$$

In finite mode the harness evaluates this finite sum and locally minimizes $|B_N(T)|^2$ by a coarse grid followed by golden-section refinement in the window around $T = \gamma/(\pi/3)$. In analytic mode it uses the exact calibrated channels

$$A(T) = L(3/2 + i\gamma, \chi), \qquad B(T) = (\pi/3)L(1/2 + i\gamma, \chi),$$

then applies the same determinant and Gram determinant checks. The reference γ table selects which displayed ordinates to validate; the focal marker remains $B = 0$ and the pencil marker remains $\det H = 0$.

20 The state of the evidence

Calibration cuts both ways. A claim stronger than its proof is a lie; but a paper that has accumulated a record and declines to weigh it misstates the evidence in the other direction. This section weighs. Nothing in it upgrades a tier: every item is a cross-reference to a statement already made above at its own strength — kernel-checked theorem or scripted measurement — and the section ends with a register of what contrary data would look like, which is what makes the weighing accountable rather than advocacy.

20.1 What the record supports

Six lines of evidence, independent in mechanism, point the same way.

(i) *[Proven] The critical abscissa is derived, not visited.* Uniform arclength placement forces the area law $r_n \sim \sqrt{n}$ (Theorem 3.9), $\sigma = \frac{1}{2}$ is the *unique* scale-balanced exponent (Theorems 3.11, 17.2), and criticality itself is chart-free midpointness — the fixed axis of the conjugation involution, with $\frac{1}{2}$ only the chart's translation constant (Section 12). The line is not a target the zeros must be argued onto; it is the only locus the geometry supports.

(ii) *[Proven] Both directions of the realization are settled; only completeness is open.* Every critical-line zero is realized as an admissible eigen-event (Theorem 15.5); every realized cancellation lies on the line (Theorem 16.2); the carrier has no native off-axis support (Corollary 17.3, Theorem 17.6). Within the model there is nowhere for an off-line zero to live: an off-line zero would be a *sourceless* zero of the 1D limit — a vanishing with no event behind it, in a coordinate the carrier does not possess (Remark 17.7).

(iii) *[Measured] The phase quantizes exactly in the fiber's own clock.* Gauged by the carrier's own radius — the proven area law acting as the clock — every cancellation cell is π to 0.001–$0.002\,\pi$ for ζ and all four tested characters; in the oscillator frame the first crossing is 0.499–$0.500\,\pi$ and every subsequent cell a half cycle, making the count the deterministic $N(T) = \lfloor \varphi/\pi + \frac{1}{2} \rfloor$ with no fluctuation term left over (Section 19.11). In the measured range there is no anomalous phase slack for a deviant configuration to hide in.

(iv) *[Measured] The medium is exactly arithmetic.* The ambient spectrum is Euler's clocks and nothing else: lines at $\ln p^k$ with the Satake weight law, conductor holes and CM combs exactly where the arithmetic puts them, exact predicted silences at the supersingular sites (Sections 19.7, 19.5). Every candidate non-arithmetic feature we found was hunted to a verdict and died: the ghost lines were window skirts, the π-multiple clocks are null with stated bounds, the close-pair carrier pattern is null (Sections 19.6, 19.13).

(v) *[Measured] The mechanism is universal, not tuned.* One unfolded amplitude curve across degree and conductor ($13.5\times$ collapse to $b(g) \approx 1.35\, g^{2.3}$, seven samples), the reopening exponent $\beta = 1.01$ of the parameter-free product law over 1517 zeros, the law holding from $g = 0.5$ down to $g = 8.2 \times 10^{-5}$, and the identical focal mechanism locating degree-two zeros with sign lanes in place of residue classes (Sections 19.9, 19.13, 19.5). Nothing is family-specific.

(vi) *[Measured] Every arch floor collapses at every vanishing the instruments have touched.* Across the record — the 111 validated Dirichlet zeros (eigenheights to 3.8×10^{15}), the 1517-zero reverb record below $t = 2000$, the degree-two firsts, the Lehmer and Gourdon pairs — nowhere does the fiber dip toward zero and reopen without vanishing where the count demands one (Sections 19.2, 19.5, 19.9). The hardest known cases are the cleanest: at the Gourdon pair ($g = 8.2 \times 10^{-5}$) the arch peaks at $\varphi = 0.500$ exactly and both ends vanish.

(vii) *[Measured, dichotomy proven] The vanishings carry the prime clocks.* Excising the dip neighborhoods removes $\approx$10% of every clock's coherent line power; by the proven dichotomy — locked positions add with J, unlocked positions cancel — the vanishing set is phase-locked to every prime clock (Section 19.10): the explicit-formula duality, observed. And the purity structure the completeness hypothesis needs is measured as an identity of the carrier metric — $C_k \leq 1$ on every vanishing class, zeros indistinguishable from random heights (Theorem 18.3).

20.2 The working hypothesis

On this record we adopt, and state as such, a *working hypothesis*: *projection completeness* — every zero has a source, equivalently GRH (Theorems 16.3, 18.7; the equivalence is per L-function, and item (v) is why we do not expect the answer to be family-dependent). It is the favored branch of the dichotomy of Remark 17.7, favored because every proven constraint and every measurement lands on it while deliberate search for the other branch's signatures (below) has produced nothing. The classical record converts under this framing: on-line $\Rightarrow$ sourced is a theorem (Theorem 15.5), so the $\sim 10^{13}$ classically verified zeros of ζ are $\sim 10^{13}$ observed *sourced* zeros — a sourceless zero has never been observed by anyone — and our instruments add the fine-grained event signatures at $\sim$1,600 of them. It is not an assertion: nothing in this paper proves it, and the other branch needs only one example. One distinction keeps the record honest in both directions: the nulls and retractions above falsified *our own structural readings*, never the sourced branch — both kinds of outcome are reported — and any future observation landing on the other branch will be published with the same prominence as any confirmation.

20.3 The falsifiability register

The hypothesis is refutable in the model's own observables. We pre-commit to the following disconfirmation signatures. Each is a concrete measurement; the first six have been deliberately sought, and the current count of each is zero; the seventh is pre-registered ahead of its first run.

(F1) *A non-collapsing arch floor.* An off-axis pair at $\frac{1}{2} \pm \delta + it_0$ floors the on-line distance product at order δ^2: an arch that dips and reopens *without* vanishing where the deterministic count of Section 19.11 demands a zero. Quantified [`tmp/offaxis_bounds.py`]: every one of the 1517 located vanishings below $t = 2000$ collapses to the taper floor (median 1.8×10^{-4}), giving per-zero displacement bounds $\delta \leq 4.5 \times 10^{-3}$ (median), $\leq 1.3 \times 10^{-2}$ (worst) — the channel fired 1517 times, including the two hardest known configurations, with zero hits.

(F2) *A genuine non-Euler spectral line*: an ambient line at a frequency outside $\{\ln p^k\}$ surviving the method law of Section 19.13 (local-maximum enumeration, span lock, blend and skirt tests, same-span controls). Two candidates arose; both were window artifacts and are retracted by name. The census closes the channel quantitatively: $81 + 79$ lines, all on $\ln p^k$; every residual peak an intermodulation $\ln p \pm \ln q$; nothing off $\log \mathbb{Q}_+^\times$; π-multiples $\leq 6.5 \times 10^{-5}$ of the $\ln 2$ power; configuration echo and long-range memory within the null (Section 19.7).

(F3) *A phase cell that is not* π: a cancellation cell in the fiber's own clock departing beyond the measured $0.002\,\pi$ floor — equivalently a height where the phase count and the vanishing count disagree. Not observed for ζ or any tested character.

(F4) *A weight-law violation*: a prime-power line whose measured weight departs from the Satake/Newton prediction beyond the measured envelope — now $\sim$0.1% with the calibrated, unclipped estimator (Section 19.10) — or a predicted exact silence that is not silent (Section 19.5). $\sim$40 lines across three forms at the older $\pm 4\%$ method precision, sixteen ζ lines at 0.1%; none.

(F5) *A universality break*: a family whose unfolded configuration statistics leave the shared curve — an amplitude exponent, reopening slope, or arch shape that is degree- or conductor-specific. Seven samples across degrees one and two; the counting statistic's number variance tracks GUE with Berry saturation on the same $\log(t/2\pi)$ clock (Section 19.8); none.

(F6) *A Castelnuovo violation*: a vanishing class whose transported cup norm grows above the half-unit rate ($C_k > 1$ for some k) — the purity engine's single open input failing where it is measurable (Theorem 18.3). Measured on 30 vanishing classes and 30 random heights, $k \leq 4$, two similitudes: $C_k \leq 1$ everywhere; none.

(F7) *An alignment-margin failure* (pre-registered; defined, not yet run). The master chain `coherence_implies_conj_axis` makes the summed-fiber HB question measurable: sample the bank at upper-half-plane points and record the per-term margin, $\|E_j^*(z)\|$ against $\mathrm{Re}\big(e^{i\theta}E_j(z)\big)$, across the bank. Persistent anti-alignment — the big-strand/small-strand conspiracy named in Remark 18.6 — would be evidence against the coherence route, and we commit to publishing the margins either way.

The correction ledger is this register's proof of operation: the machinery has fired repeatedly, on us — the growth-clock retraction and the ghost lines (Section 19.13), the conductor-hypothesis inversion (Section 19.12), the ACF first reading (Section 19.7), and the clip-bias self-catch that turned a 1% anomaly into a method law and a sharper result (Section 19.10) — and every firing was published rather than suppressed, several upgrading into results. The instruments and their commands are public (Sections 19.14, 23); the reader is invited to run them and weigh independently.

21 A Langlands reading

This section is interpretation: a reading of the proved and measured structure above in Langlands vocabulary. Nothing in it is a theorem or a measurement beyond the cross-references given; it is recorded because it organizes the phenomenology and names the experiments it suggests.

The helix is read as a *lifted representation* of the L-function: the one-dimensional Dirichlet L-function is the *character* of the three-dimensional phasor representation — projection as trace

— and the ambient spectroscopy of Section 19.7 is then the recovery of a representation from its character, strong multiplicity one enacted empirically. The dual group appears as *the clock-face of each prime*: for GL(1), characters of $\mathbb{C}^\times$ — the root-of-unity cells of Theorem 3.15; for GL(2), the conjugate Satake pair (the transverse conjugate-pair block of Theorem 14.4, made local at p), with the p-power ladders as SU(2) characters and cell closure passing to Sato–Tate orthogonality. On this reading, the Satake weight law of Section 19.5 — exact to $\sim$0.1% once the estimator is calibrated and unclipped (Section 19.10) — is a direct measurement of the Satake isomorphism. The organs align: the height axis is the archimedean component; the Frobenius similitude's radial factor $\sqrt{m}$ (Theorem 14.2) supplies the weight/purity normalization that étale cohomology provides over $\mathbb{F}_q$, here forced by the carrier's area law; the $\pi/3$ cell is a level/cyclotomic trivialization — gauge in the readout (provably inert in the arrows, Section 19.5), structural in the carrier. The reading is falsifiable in the model's own terms: functoriality becomes a spectral identity between mediums — and the $\mathrm{Sym}^2\Delta$ test is now run and two-sidedly confirmed (Section 19.8) — the generalized Ramanujan bound becomes a ladder decay rate, and reduction types of elliptic curves are already spectroscopically distinct (the conductor holes and CM combs of Sections 19.7 and 19.5). The root number reading — the weld phase of the two conjugate strands at the hinge, with $\varepsilon = -1$ forcing a hinge zero and the hinge reopening rate tied to the leading central datum — is likewise interpretation with experiments pending.

22 Structural frameworks (outlook)

Three machine-checked frameworks sit above the main line. Each is correct and kernel-checked; they are collected here as the platform for the next stage of the program.

22.1 The chiral cup

The forward/reverse chiralities of the double-ended carrier assemble a Hermitian, positive *definite* sesquilinear cup on the finitely-supported fibre space, which completes to a genuine Hilbert space with the ℓ^2 energy on the diagonal (`cup_hermitian`, `cup_null_iff`, `hilbert_completion_exists`); the Frobenius transport acts as a cup similitude scaling by $\|\mu\|^2$, unitary after normalization (`transport_unitary_after_normalization`), a crossing-induction argument propagates chiral modulus balance (`no_dominance_by_crossing_induction`), and on the prime-power probe the cup energy carries the von Mangoldt field, $-L'/L$ for $\mathrm{Re}\, s > 1$ (`cup_intertwining`, `vonMangoldtChannel_eq_neg_logDeriv`).

22.2 The Kähler polarization and the trace

The fibre carries a compatible triple (Ω, J, g) with the Kähler identity $\langle x, y\rangle = g(x, y) + i\,\Omega(x, y)$, strictly positive on the chiral defect (`inner_eq_gH_add_I_OmegaH`, `theoremA`); Frobenius transport scales the symplectic pairing by exactly p^r (`theoremB`), and the weighted local trace of the site-shift dynamics is the von Mangoldt superposition, equal to $-L'/L$ for $\sigma > 1$ (`theoremC`). A cohomological chart of the vanishing is fixed ($H^\bullet(\rho) = \mathbb{C}/\langle L(\rho, \chi)\rangle$; `theoremD`): as formalized it is the field-theoretic identity "$\mathbb{C}/\langle x\rangle \neq 0 \iff x = 0$" — the dictionary entry that fixes where a genuine cohomology theory must land.

22.3 The de Branges evaluation

The Hermite–Biehler/de Branges framework is formalized generically: positivity of the structure kernel forces the HB inequality and hence a real, discrete spectrum (`featureMap_forces_HB`, `hb_no_zero_upper`, `Bcomp_zeros_discrete`), with the Paley–Wiener function e^{-iz} as the fully unconditional worked template (`paleyWiener_isHB`; spectrum $\{k\pi\}$).

Corollary 22.1 (On-line cancellations are real spectral points; `criticalLine_deBranges_real`, `zeroHarmonic_deBranges_real`)**.** *Under the change of variables $z = -i(\rho - \frac{1}{2})$, a carrier vanishing at height γ sits at the* real *de Branges point $z = \gamma$; reality of z is exactly $\operatorname{Re}\rho = \frac{1}{2}$ (`deBranges_var_im`).*

Remark 22.2 (Scope: IsHB Λ is classical open work, not used here)**.** *That a Λ-built structure function is Hermite–Biehler is equivalent to the Riemann Hypothesis — the classical de Branges reformulation. No such structure function is constructed here; only the unconditional on-line content (Corollary 22.1) is used. The open question of this paper is projection completeness (Remark 17.7, Theorem 18.7), not* IsHB Λ *— though the carrier's own clocks are now proven to be real de Branges functions on the HB boundary (Theorem 18.5), giving the frame a geometric foothold; the remaining wall is a single implication away (Remark 18.6).*

23 Formalization

Every definition and theorem above is formalized in Lean 4 [19] against Mathlib [18], in the package `RequestProject` (files `ClosedForm.lean`, `HelixLogFreeFTA.lean`, `HelixCollapseReality.lean`, `LFunctionPhasor.lean`, `GeometricProjectionHolds.lean`, `Faithfulness.lean`, `AreaLaw.lean`, `UnconditionalFrobenius.lean`, `DeBranges.lean`, `FrobeniusSimilitude.lean`, `PerHeightConvergence.lean`, `HeightGrowthActive.lean`, `CircleMonodromy.lean`, `CriticalLineBridge.lean`, `GeometricPhasorClosure.lea` `Phasor3D.lean`, `FocalEigenheight.lean`, `HarmonicPencilCell.lean`, `FinitePencil.lean`, `AdmissibleEigenheight.lean`, `FullFiber.lean`, `PrimePowerProbe.lean`, `CupIdentity.lean`, `ChiralCup.lean`, `CrossingInduction.lean`, `CrossingEncodingAudit.lean`, `HelixPolarization.lean`, `ConditionalToUnconditional.lean`, `SpectralFiberIsLFunction.lean`, `SpectralZeroKernel.lean`, `CancellationEquivalence.lean`, `SelfAdjointGeneratorReadout.lean`, `SpectralSignFlip.lean`, `NoDoubleCancellation.lean`, `SpectralExhaustion.lean`, `SpectralGramEvent.lean`, `SpectralRealizationGR` `FocalResidualVanishes.lean`, `FocalCancellationFindsZeros.lean`, `FocalCancellationEigenstate.lean`, `GeometricReadout.lean`, `ReverbResidue.lean`, `GUEBridge.lean`, `MellinDual.lean`, `UnitMidpoint.lean`, `HingeKernel.lean`, `AntihelixWindow.lean`, `WeilDuality.lean`, `ChiralityHB.lean`, `BSDLadder.lean`, `ProjectionCompleteness.lean`, `ClockDipDuality.lean`, `DirichletDuality.lean`, `SummedFiberHB.lean`, and `MeromorphicTraceIdentity.lean`). Each named theorem compiles with no `sorry`; the kernel axiom footprint reported by `#print axioms` (equivalently `lean_verify`) is

$$\{\ \text{propext},\quad \text{Classical.choice},\quad \text{Quot.sound}\ \},$$

the three standard Mathlib axioms, with no construction-specific axiom. The correspondence between the statements here and the Lean declarations is given by the identifiers in parentheses throughout.

The capstone identities — the strip extension (Theorems 7.1, 7.2, 7.3), the conjugacy of the two chiralities (Theorem 11.1), and the determinant identity (Theorem 14.3) — are collected in `Faithfulness.lean`; the geometry (Section 3) and the log-free winding (Section 4) live in `ClosedForm.lean` and `HelixLogFreeFTA.lean`, with the emergent area-law radius (Theorem 3.9, Definition 3.8) and the scale-critical exponent (Theorem 3.11) in `AreaLaw.lean`. The self-adjoint eigenstate design (Section 14.1) lives in `UnconditionalFrobenius.lean` and the Hermite–Biehler / de Branges framework (Section 22.3) in `DeBranges.lean`, with their connection to the carrier — the eigenstate identification (`spin_eq_spectralWave`, `frobenius_eigenstate_det_one`) and the on-line de Branges evaluation (`criticalLine_deBranges_real`, `zeroHarmonic_deBranges_real`) — collected in `Faithfulness.lean`. The compact local spectral mechanism of Section 14 — the Frobenius similitude (Definition 14.1, Theorem 14.2), the unitary transverse block (Theorem 14.4), the eigenstate–data orthogonality (Proposition 14.8), and the admissibility / no-off-axis-support results of Section 17 (Theorems 17.2, 17.5, 17.6) — lives in `FrobeniusSimilitude.lean`, assembled in `frobenius_local_spectral_mechanism` and `no_native_offaxis_support`. The remaining layers map as follows: the per-height convergence on the line (Section 7.1) is `PerHeightConvergence.lean`; the constant height law and active fraction (Remark 3.2) and the exact $\pi/3$ monodromy (Theorem 3.15) are `HeightGrowthActive.lean` and `CircleMonodromy.lean`; the geometric phasor-closure dictionary and lane split (Section 7) are `GeometricPhasorClosure.lean` and `CriticalLineBridge.lean`; the three-channel 3D phasor (Section 6) is `Phasor3D.lean`; and the harmonic pencil, focal eigenheight, and source-height admissibility (Section 15) are `HarmonicPencilCell.lean`, `FocalEigenheight.lean`, and `AdmissibleEigenheight.lean`. The chiral cup and its Hilbert completion (Section 22.1) live in `CupIdentity.lean`, `ChiralCup.lean`, `CrossingInduction.lean` (with the non-vacuity audit `CrossingEncodingAudit.lean`) and `PrimePowerProbe.lean`; the Kähler polarization, the trace formula, and the cohomological re-expression of the zeros (Section 22.2) are `HelixPolarization.lean`; the convergent η-fibre (Section 7.2) is `FullFiber.lean`; and the unconditional Paley–Wiener template (Section 22.3) is `ConditionalToUnconditional.lean`.

The updated spectral/focal layer is split into smaller audit modules. `SpectralFiberIsLFunction.lean` proves that the 3D spectral-fiber readout converges to the analytic L-function (`spectral_fiber_is_Lfunction`) and that the eta-configured principal readout recovers ζ on the punctured strip (`etaSpectralFiber_readout_recov` `SpectralZeroKernel.lean` and `CancellationEquivalence.lean` package zeros as kernel events of the configured spectral operator (`spectral_zero_has_kernel`, `spectral_kernel_iff_zero`, `cancellation_equivalence`). The self-adjoint readout and its Gram markers live in `SelfAdjointGeneratorReadout.lean`, `SpectralSignFlip.lean`, `NoDoubleCancellation.lean`, `SpectralExhaustion.lean`, and `SpectralGramEvent.lean`: these record the spectral height `specHeight`, the signed mode `specBchan`, the rank-drop criterion `specGram_det_zero_iff`, sign-change and real-axis facts (`spectral_cancellation_sign_flip`, `sign_flip_only_on_real_axis`), distinct-mode exclusion (`no_double_cancellation`), and the critical-line implication from an L-zero to a harmonic Gram event (`L_zero_imp_spectral_gram_event`). The focal-cancellation modules `FocalResidualVanishes.lean`, `FocalCancellationFindsZeros.lean`, and `FocalCancellationEigenstate.lean` formalize that the normalized focal residual vanishes exactly with the scalar L-readout (`focal_residual_zero_iff_L_zero`), that an on-line zero at ordinate y is found at source height e^y (`focal_cancellation_finds_online_zero`), and that an exact charged focal cancellation produces a standing/eigenstate package with conserved magnitude (`focal_cancellation_produces_eigenstate`). `GeometricReadout.lean` collects the height-readout and no-double-residue facts, and `MeromorphicTraceIdentity.lean` records the completed trace identities for the spectral operator. `ReverbResidue.lean` and `GUEBridge.lean` formalize the reopening/cluster kernels, the CUE bridge, the cell-closure arithmetic, and the Satake silence of Sections 19.9 and 19.5; `UnitMidpoint.lean` and `MellinDual.lean` carry the midpoint charts and the self-dual readout of Section 12. The 2026-07 arc adds seven modules: `HingeKernel.lean` (the hinge turning point, weld-ray pinning, jet parity — Section 13); `AntihelixWindow.lean` (the incomplete-Γ growth window, determinant-one strand weights, the term-local weld — Theorem 13.4); `BSDLadder.lean` (the hinge dimension and its forced parity, the G_r reference ladder — Theorems 13.3, 13.5); `WeilDuality.lean` (the duality pairing, the purity engine, the Castelnuovo reduction, the membership boundary — Theorems 18.1–18.4); `ChiralityHB.lean` (the purity-defect law and clock-RH — Theorem 18.5); `ProjectionCompleteness.lean` (the frame split — Theorem 18.7); and `ClockDipDuality.lean` (the clock–dip inference dichotomy — Section 19.10); `DirichletDuality.lean` then carries the duality package to every conductor (Section 18), and `SummedFiberHB.lean` holds the coherence kernels and the master chain (Remark 18.6).

Reproduction. The Lean claims: `lake build` (Mathlib pinned by the manifest); axiom footprints via `#print axioms`. The phenomenology of Section 19: `python3 model.py` (absorption law, evaluator-vs-brute on 240 configurations, pencil family), `python3 focal_closure.py test` (growth locator; the degree-two families with Hecke/Deligne build-time checks), and `python3 carrier_fiber.py test` (finite carrier-clock harness). The interactive explorer (`phasor_explorer/explorer_v2.html`) exposes every character, every listed zero, and every magnitude law of this paper. Appendix A is the claim-by-claim map.

Remark 23.1 (Scope)**.** *This paper presents a geometric state space: the double-ended Frobenius helix carrier and its phasor fiber, whose admissible scale-balanced states exist only on the critical line (Section 17) and on which the vanishing points are* exactly *the on-line zeros of ζ and every Dirichlet L-function, by the formalized equivalence (Theorem 10.1, Section 7.2). Its central contribution is intrinsic: the carrier has* no native off-axis support *(Theorem 17.6), and RH is reframed as* every zero has a source *— no sourceless zeros in the 1D limit (`everyZeroHasSource_iff_critical`), equivalent to GRH, with the off-axis branch proved impossible inside the model. Its mathematical content is exactly the statements proved above — the carrier, the area-law derivation of the critical exponent (Theorem 3.11), the accumulation to $L(s,\chi)$ on $\operatorname{Re} s > 0$, the explicit-formula reading of the zeros as poles, the reality of the completed object on the critical line, the conjugacy of the two chiralities, the Frobenius similitude (Theorem 14.2), the unitary transverse block (Theorem 14.4), the eigenstate–data orthogonality (Proposition 14.8), the self-adjoint eigenstate realization of the chiral eigenphases (Section 14.1), the de Branges evaluation placing the on-line cancellations as real spectral points (Section 22.3), and the admissibility restriction of the carrier's state space to the critical line (Section 17). The pivotal hypothesis is* projection completeness *(Remark 17.7, Theorem 18.7) — formalized in three guises: that every nontrivial zero arises from a real source height (`EveryZeroHasSource`, `everyZeroHasSource_iff_critical`), that every zero of the completed continuation arises from a helix vanishing (`projection_complete_iff_RH`), and that a vanishing* produces *the eigenstate (Remark 14.11). The classical de Branges* IsHB Λ *condition (Remark 22.2) is the same RH content in the de Branges form — connected here but neither formalized nor relied on (open work, not a conditional of this paper). The work poses the dichotomy — every zero has a source, so all lie on the line,* or *the 1D limit admits a* sourceless *zero, in a coordinate the carrier does not possess — proves that the off-axis branch cannot arise within the 3D carrier, and weighs the evidence on the 1D question (Section 20). The Riemann Hypothesis is exactly the resolution of this dichotomy; it is neither assumed nor proved here.*

A Headline claims and their Lean identifiers

Each row is a claim of this paper, the Lean identifier that proves it, and its file (package `RequestProject`, 55 modules). All rows compile with no `sorry`; axiom footprint $\{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$.

Claim	Lean identifier	File
Area law $r_n^2/n \to r\Delta/\pi$	`windIntegerSite_radius_sq_tendsto`	`AreaLaw`
$\sigma = \frac{1}{2}$ uniquely scale-balanced	`sigma_half_is_scale_critical`, `scaleBalanced_iff`	`AreaLaw`
$\pi/3$ has exact finite monodromy	`pi3_finite_monodromy`	`CircleMonodromy`
Strip convergence to $L(s,\chi)$, $\mathrm{Re}\, s > 0$	`dirichlet_strip_tendsto_LFunction`	`LFunctionPhasor`
η-channel identifies ζ's on-line zeros	`FullFiber_eq_zero_iff_zeta`	`FullFiber`
Vanishing = eigenwave–data orthogonality	`fiberCarrierFinite_eq_zero_iff_orthogonal`	`FrobeniusSimilitude`
Channel closure $\iff$ L-zero	`channel_closure_iff_L_zero`	`FocalEigenheight`
Pencil det $= (\lambda - \mu)AB$; rank-drop $\iff$ L-zero	`harmonicPencil_det`, `gramH_rank_drop_iff_L_zero`	`HarmonicPencilCell`
Rank-drop is calibration-independent	`gram_rank_drop_calibration_independent`	`HarmonicPencilCell`
Channel-agnostic detector (no L in statement)	`harmonicGram_rank_drop_iff_channel_zero`	`HarmonicPencilCell`
Finite admissibility by positivity	`finiteA_pos`, `finite_gramH_rank_drop_iff_channel_zero`	`HarmonicPencilCell`
Finite pencil $\to L$ only at final verification	`finite_pencil_rank_drop_iff_L_zero`	`FinitePencil`
Every critical-line zero realized at $Z = e^\gamma$	`nontrivial_zero_represented`	`HarmonicPencilCell`
One cancellation site per event	`no_double_cancellation`	`NoDoubleCancellation`
Realized cancellation $\Rightarrow$ on the line	`nontrivial_A`χ`_kernel_no_offline`	`SpectralRealizationGRH`
Realization of all cancellations $\Rightarrow$ GRH	`GRH_from_nontrivial_A`χ`_realization`	`SpectralRealizationGRH`
Every zero has a source $\iff$ critical line (= GRH)	`everyZeroHasSource_iff_critical`	`HarmonicPencilCell`
No native off-axis support	`no_native_offaxis_support`	`FrobeniusSimilitude`
Reality of Λ on the line	`completedRiemannZeta_critical_line_im_zero`	`HelixCollapseReality`
Reopening law; residue $\iff$ simple; cluster product	`reopening_slope`, `residue_exists_iff_simple`, `cluster_product_law_norm`	`ReverbResidue`
RMT derivative statistic = distance product	`cue_rate_eq_distance_product`, `cue_reopening_slope`	`GUEBridge`
Step closes $\iff$ $c \in 2\pi\mathbb{Q}$; integer steps never	`step_closes_iff`, `integer_step_never_closes`	`GUEBridge`
Satake silence: p^2-line dead $\iff$ $\lambda^2 = 2$; E11 at p=2	`satake_silence_iff`, `e11_ln4_silent`	`GUEBridge`
Criticality is midpointness; $\frac{1}{2}$ a chart constant	`riemann_chart`, `centered_chart`	`UnitMidpoint`
UNIT/2 in every gauge; $\pi/3$ cell reflects about $\pi/6$	`criticality_is_half_unit`, `eisenstein_conj_axis`	`UnitMidpoint`
Prime clocks never collide (unique factorization)	`prime_clocks_incommensurable`	`UnitMidpoint`
Readout self-dual: spectral axis = height axis	`readout_dual_recovers_sites`	`MellinDual`
Hinge is a turning point (even collapse wave)	`collapseWave_even`, `hinge_turning_point'`	`HingeKernel`
Weld ray = $\arg(\varepsilon)/2$; ε=-1 forces hinge zero	`weld_pins_half_phase`, `weld_minus_one_forces_zero`	`HingeKernel`

Claim	Lean identifier	File
Jet parity: $(-1)^{d(0)} = \varepsilon$	`hingeDim_parity`, `even_live_jet_is_even`, `odd_live_jet_is_odd`	`BSDLadder`, `HingeKernel`
Fricke midpoint = UNIT/2, conductor as unit	`fricke_midpoint_is_half_conductor_unit`	`HingeKernel`
Antihelix window; det-1 strand weights; term-local weld	`gamma_splits_at_cut`, `strand_weights_det_one`, `weld_kills_each_phasor`	`AntihelixWindow`
Reference ladder positive; G_1 = reverb kernel	`Gr_nonneg`, `Gr_one_eq`	`BSDLadder`
Duality: perfect det-1 pairing, $d(\rho) = d(1-\rho)$	`vanishing_dual_pair`, `dual_dimension_symmetry`, `dual_pair_det_one`	`WeilDuality`
Purity engine; Castelnuovo reduction	`tensor_power_purity`, `duality_forces_purity`, `purity_from_castelnuovo`	`WeilDuality`
Conjugation axis = membership boundary of the completion	`conj_axis_is_membership_boundary`	`WeilDuality`
Purity-defect law; clock-RH unconditional	`clock_zero_depth`, `carrier_zeros_real`	`ChiralityHB`
The clock is a real de Branges function (HB boundary)	`symClock_star`, `symClock_selfdual_modulus`	`ChiralityHB`
Frame split: RH $\iff$ projection complete; 3D capstone	`projection_complete_iff_RH`, `helix3D_RH`	`ProjectionCompleteness`
Clock–dip dichotomy (position = phase; locked adds, unlocked cancels)	`dip_projection_phase`, `locked_dips_add`, `unlocked_dips_cancel`	`ClockDipDuality`
Duality at every conductor: pairing swaps χ; $d_\chi(1-\rho) = d_{\chi^{-1}}(\rho)$	`vanishing_dual_pair_dirichlet`, `dual_dimension_symmetry_dirichlet`	`DirichletDuality`
Coherence kernels; master chain: coherence $\Rightarrow$ conjugation axis	`aligned_strict_sum_HB`, `coherence_implies_conj_axis`	`SummedFiberHB`

References

[1] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsberichte der Berliner Akademie, 1859.

[2] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974.

[3] E. C. Titchmarsh, *The Theory of the Riemann Zeta-function*, 2nd ed. (rev. Heath-Brown), Oxford University Press, 1986.

[4] H. Davenport, *Multiplicative Number Theory*, 3rd ed., Springer GTM 74, 2000.

[5] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, AMS Colloquium Publications 53, 2004.

[6] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer Undergraduate Texts in Mathematics, 1976.

[7] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*, Cambridge Studies in Advanced Mathematics 97, 2007.

[8] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Springer GTM 84, 1990.

[9] L. V. Ahlfors, *Complex Analysis*, 3rd ed., McGraw-Hill, 1979.

[10] B. Ya. Levin, *Distribution of Zeros of Entire Functions*, rev. ed., Translations of Mathematical Monographs 5, American Mathematical Society, 1980.

[11] L. de Branges, *Hilbert Spaces of Entire Functions*, Prentice-Hall, Englewood Cliffs, N.J., 1968.

[12] J. T. Tate, *Fourier analysis in number fields and Hecke's zeta-functions*, in J. W. S. Cassels and A. Fröhlich (eds.), *Algebraic Number Theory*, Academic Press, 1967, pp. 305–347.

[13] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, in *Analytic Number Theory* (Proc. Sympos. Pure Math. XXIV, St. Louis, 1972), American Mathematical Society, 1973, pp. 181–193.

[14] M. V. Berry and J. P. Keating, *The Riemann zeros and eigenvalue asymptotics*, SIAM Review **41** (1999), 236–266.

[15] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N.S.) **5** (1999), 29–106.

[16] P. Deligne, *La conjecture de Weil. I*, Publ. Math. IHÉS **43** (1974), 273–307.

[17] The LMFDB Collaboration, *The L-functions and Modular Forms Database*, `https://www.lmfdb.org`, 2026.

[18] The mathlib Community, *The Lean Mathematical Library*, Proceedings of CPP 2020, ACM, 2020.

[19] L. de Moura and S. Ullrich, *The Lean 4 Theorem Prover and Programming Language*, Proceedings of CADE-28, Springer LNCS 12699, 2021.